

Color

(or why Jet sucks)

Sarah Groves



<https://www.nationalgeographic.com/photography/article/photo-of-the-day-may-2026-nomads-canyons-dragonflies>

Obesity in the US, 2008

These are two different ways of looking at the same attribute: obesity rate by state.

1. [6pt] In both charts, the attribute obesity rate is encoded by the same channel.

(a) What is this channel? (1-2 words, as specific as possible)

(b) What is the domain? [min, max]

(c) What is the range? [min, max] (you can use names instead of a quantitative range)

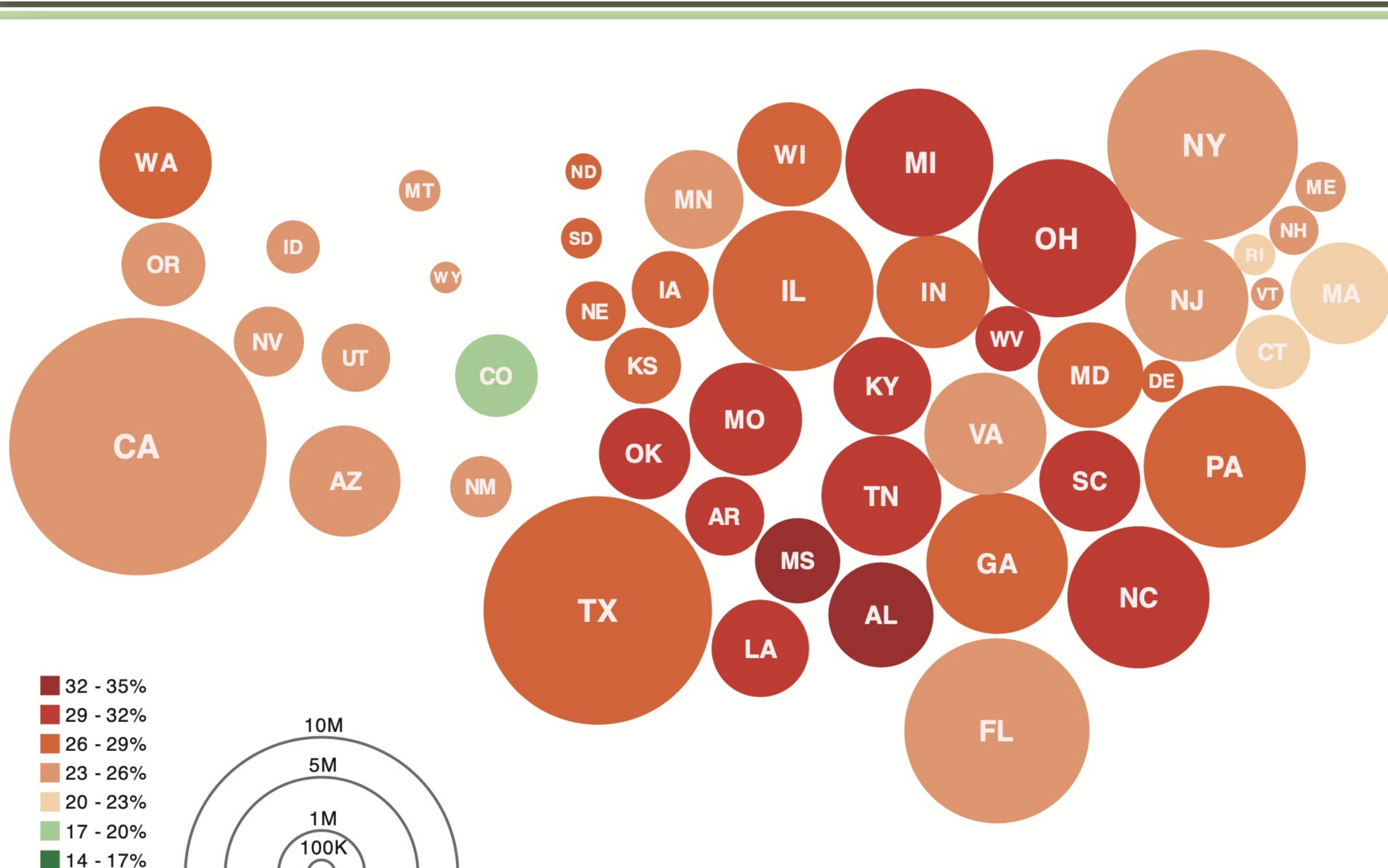
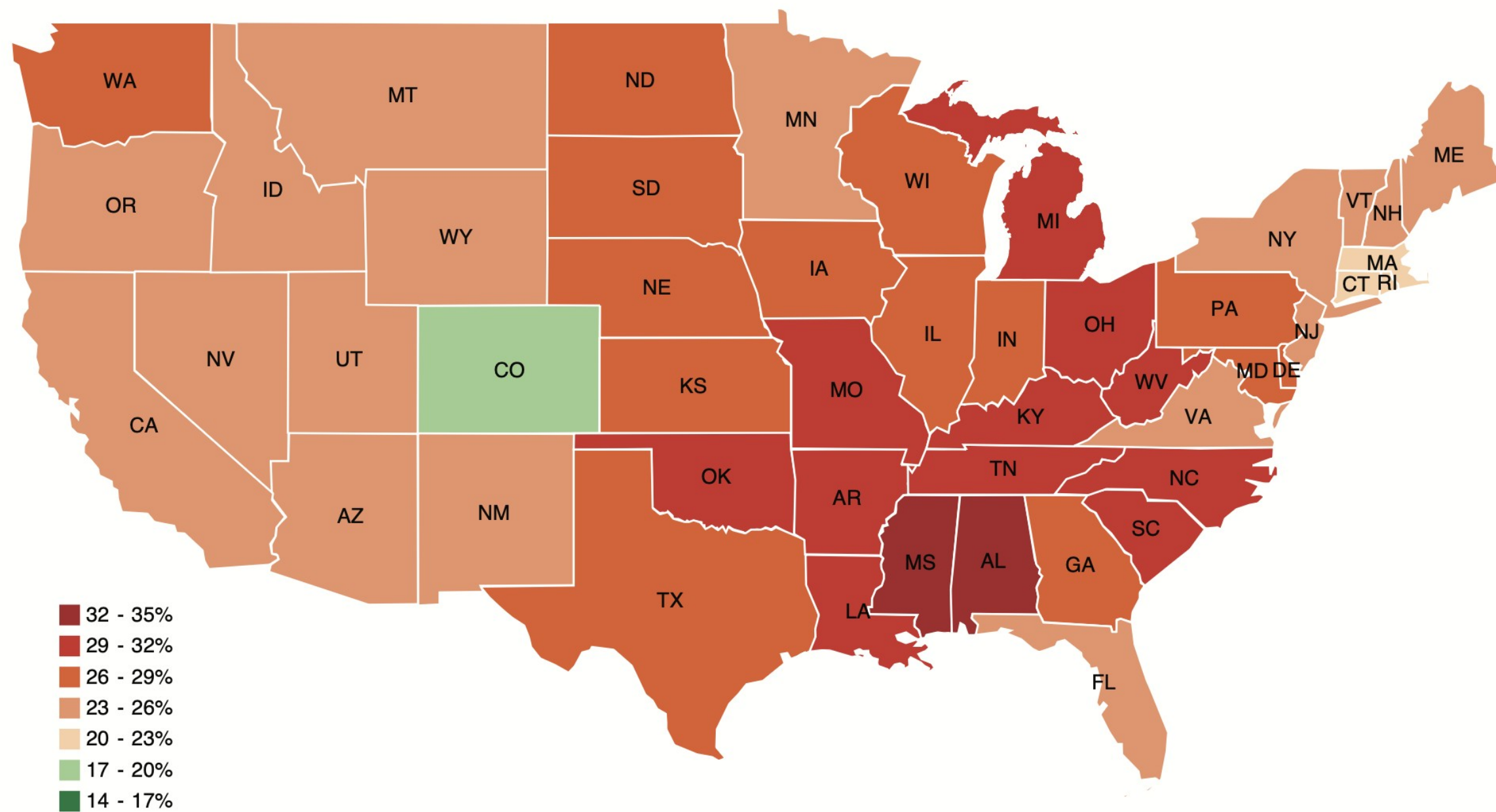
2. [4pt] In the top map,

State name (N) → spatial position

In the bottom map,

population size (NOIR?) → what channel? (1-2 words)

Bonus (+2): What is the name of the type of map in the top chart?



Logistics/due dates

- End of unit 1 **survey** (due **Sep 19**)
- **Participation grades** for unit 1: look at comments!
- Add to the **discussion board** on Vis Zoo if you haven't already!
- Today's **in class activity** will be turned in on Canvas by **Friday**
- **Project 1**: look at my comments about the dataset you chose
 - If you need to make changes, do so and resubmit under checkpoint #1
 - If you are ready, add 3 questions for **checkpoint #2** (due **Sep 27**)

What you will learn today

- How do our eyes perceive color?
- How do we represent color numerically?
- Principles of Color in Data Visualization
- What colormaps you should use
- Should we use multiple visual channels for a single attribute?

Why does this matter?

- Data visualization should be accessible (color blindness)
- Bad color choices can mislead

Design Choices Make a Difference

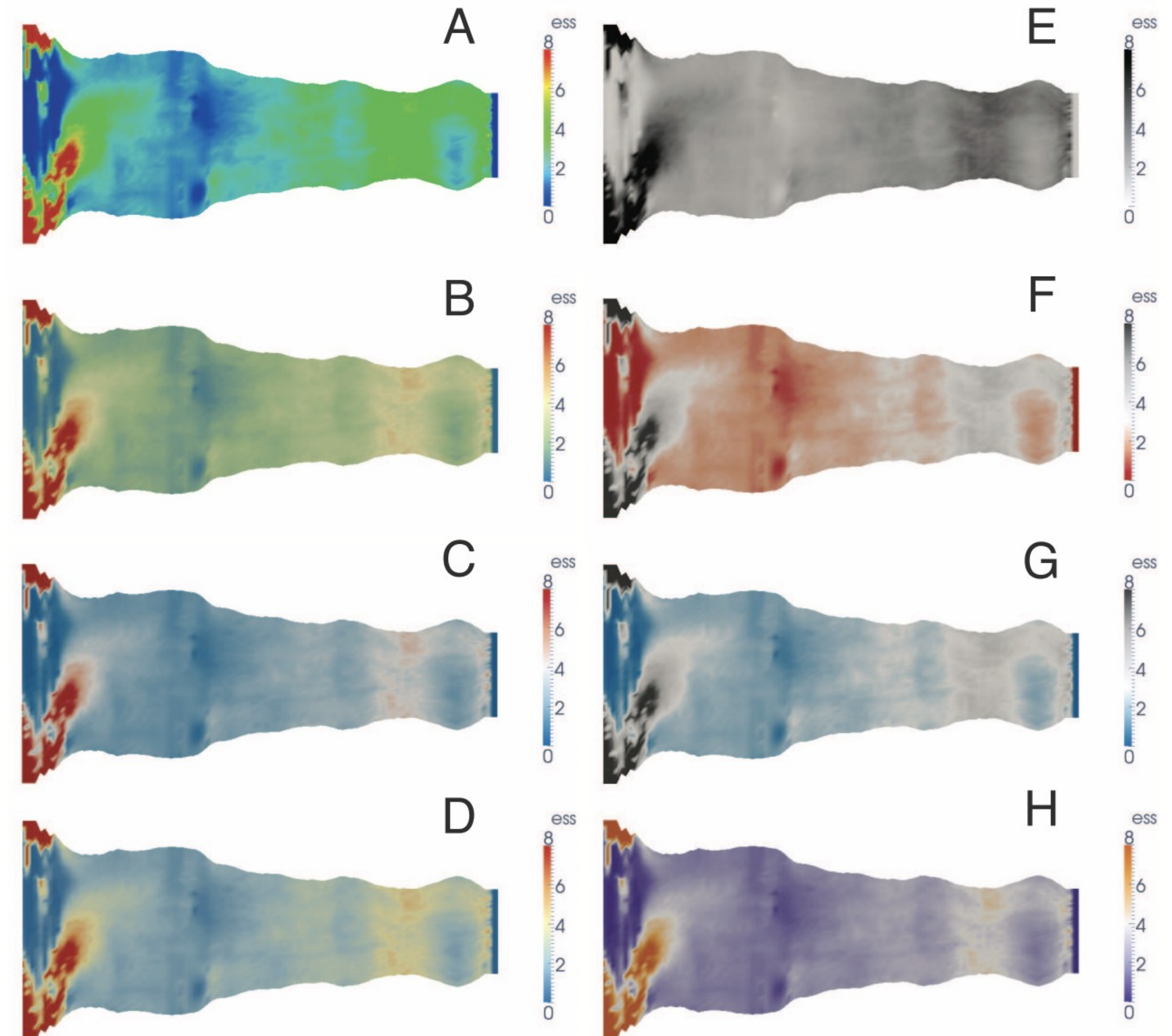
Borkin et al. Artery Visualization

You are a **doctor** who looks at cardiovascular imaging to diagnose heart disease.

The image shows a patient's artery, colored by **Endothelial Shear Stress** (force of blood pushing against the arterial walls); **low ESS** is a risk factor for heart disease (**low # = bad**).

Which colormap do you think would make it easiest to identify a patient at risk?

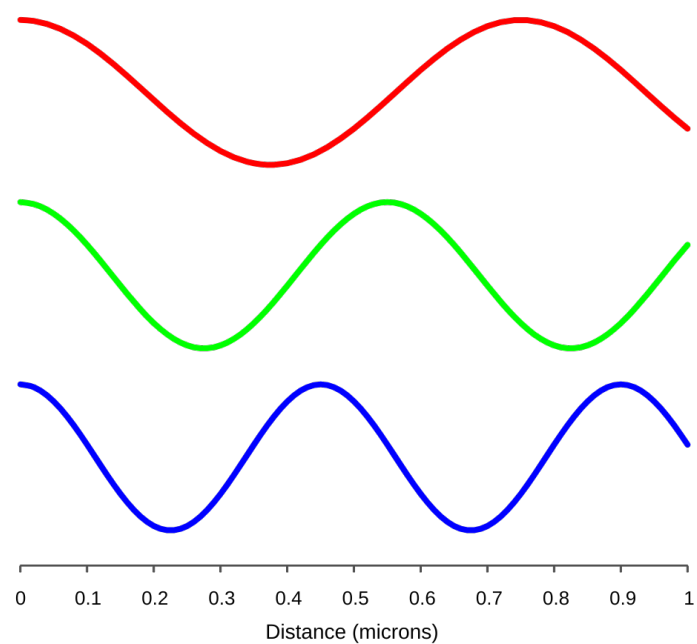
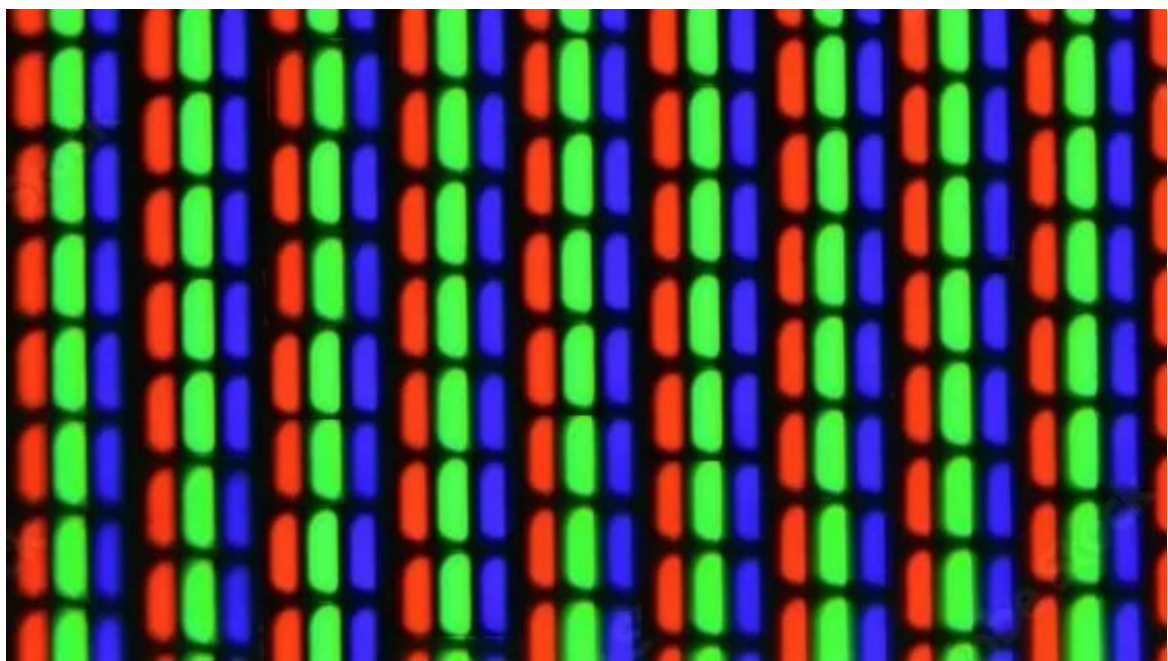
Same artery, different color maps



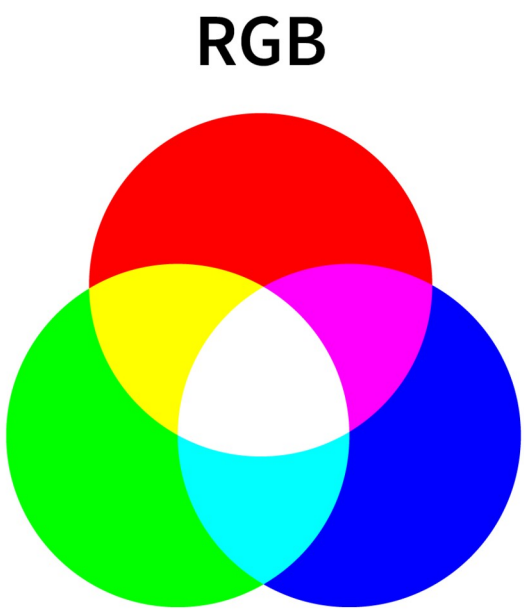
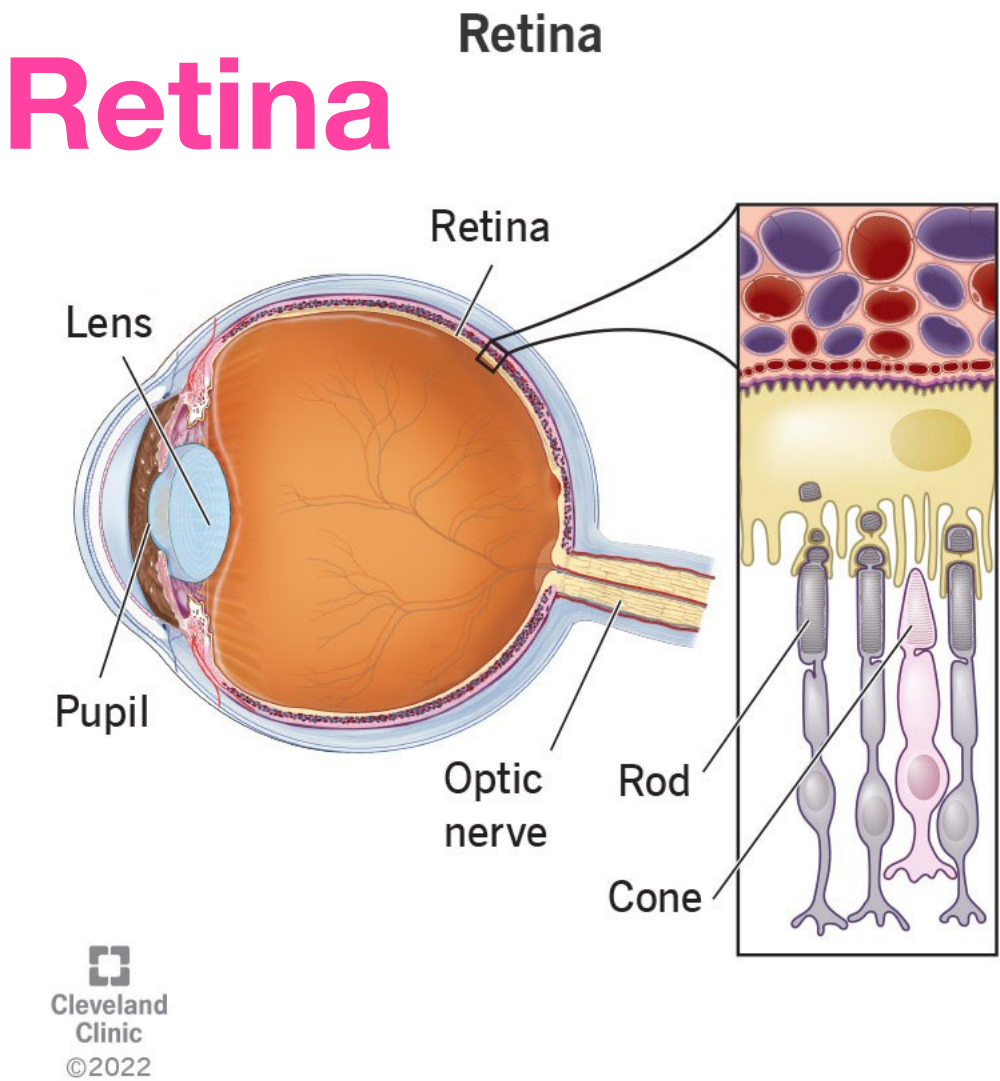
How do our eyes perceive color?

A short introduction to color theory and perception (more on perception next week!)

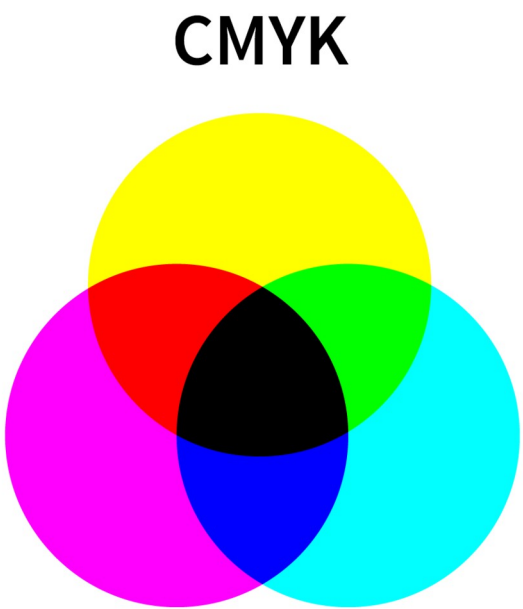
Monitor



Light



RGB



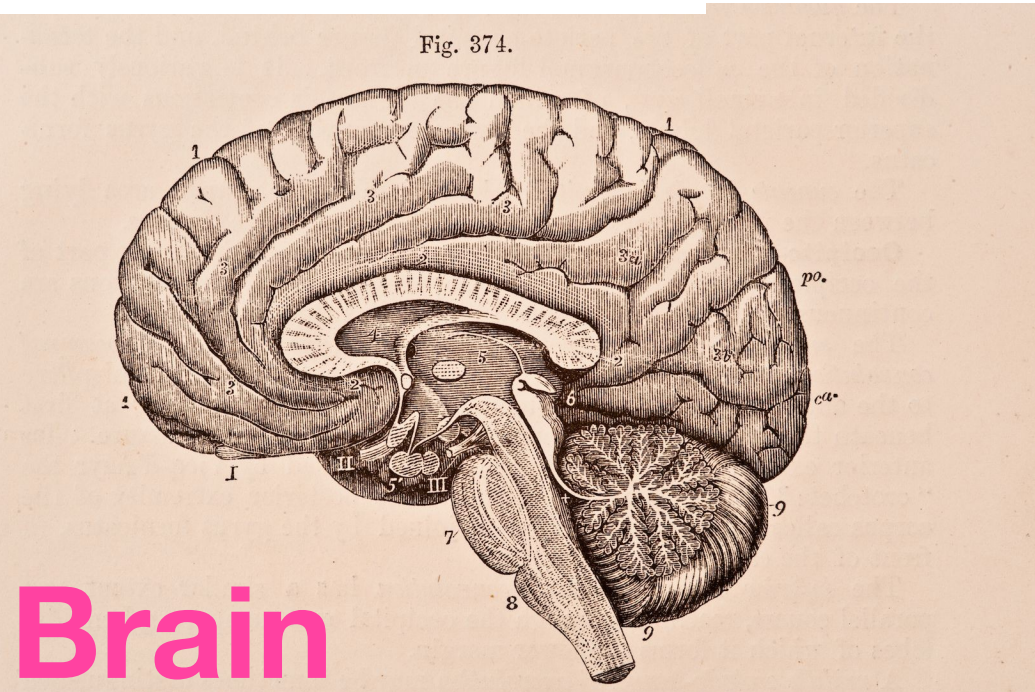
CMYK

© TechTerms.com

RGB

DATA

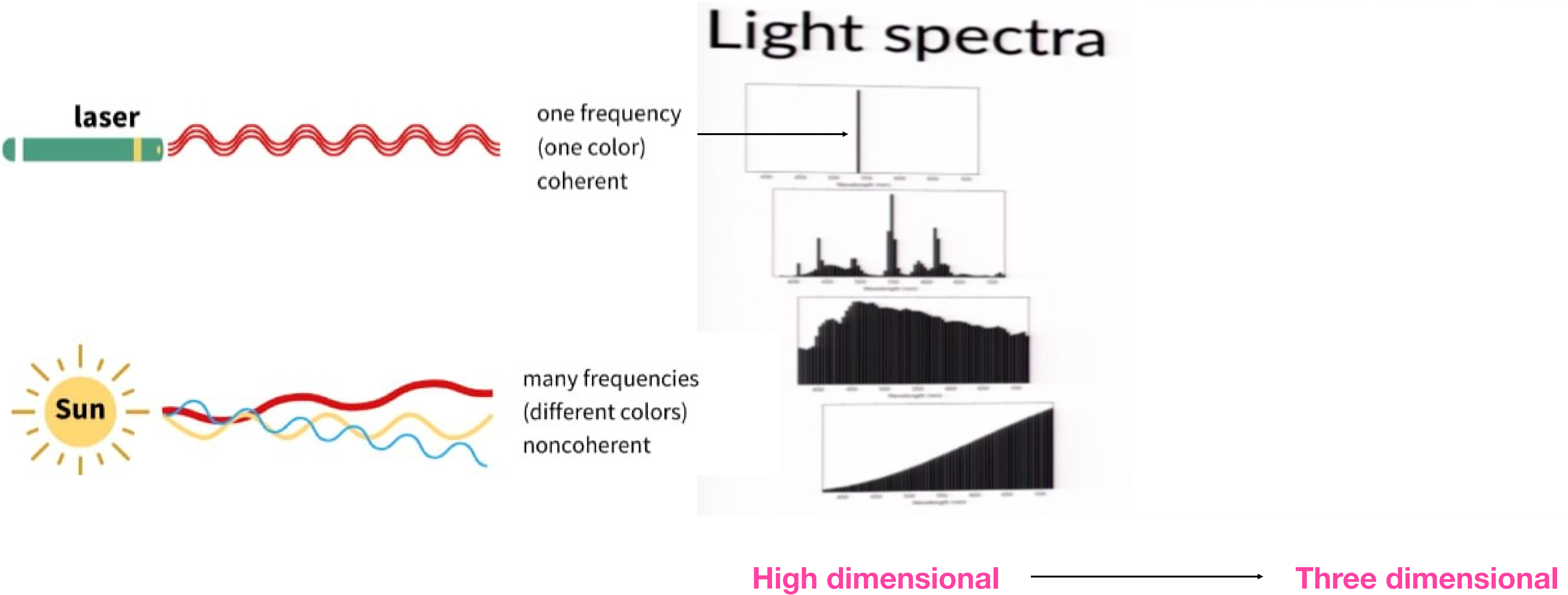
1,527	51.6
1,477	50.2
6,000	50.7
2,531	50.3
3,469	51
1,000	50.2



Brain

PERCEPTION

High-dimensional light spectra → 3 cone cells



COLOR BLINDNESS

10% of males, 1% of females (probably due to X-chromosomal recessive inheritance)

Most common: red-green weakness / blindness

Reason: lack of medium or long wavelength receptors, or altered spectral sensitivity (most common: green shift)



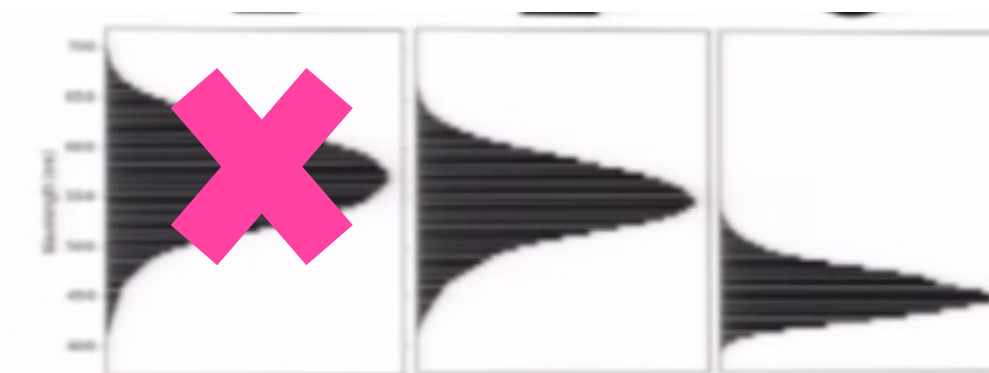
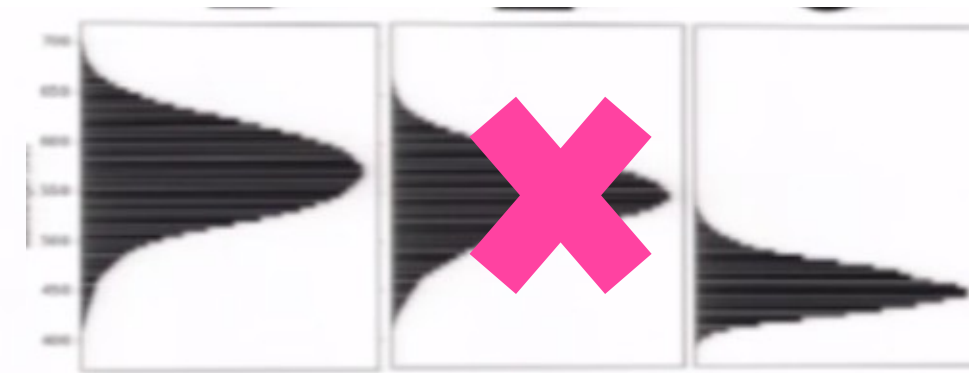
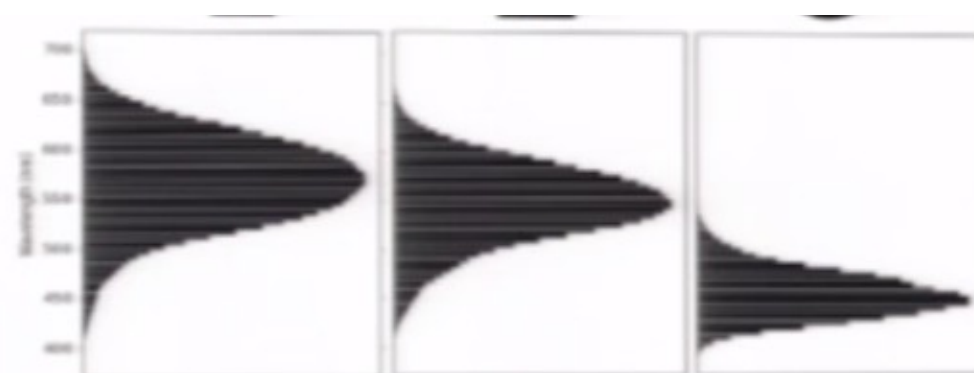
Normal Color Perception



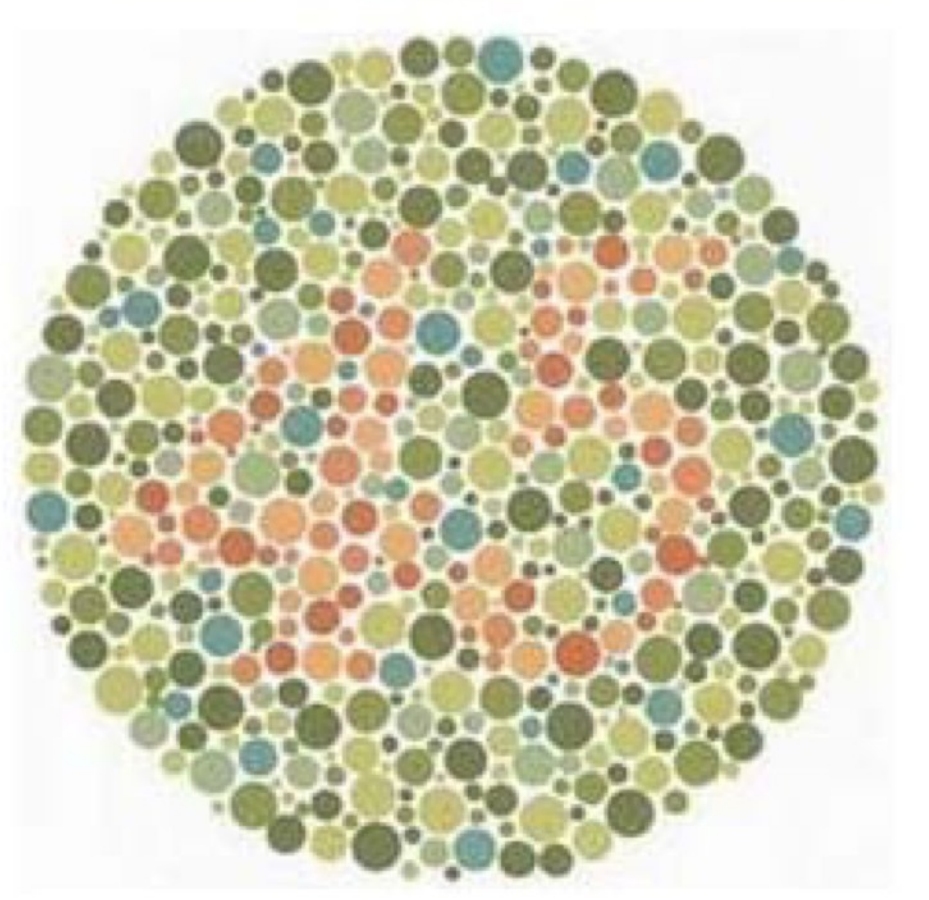
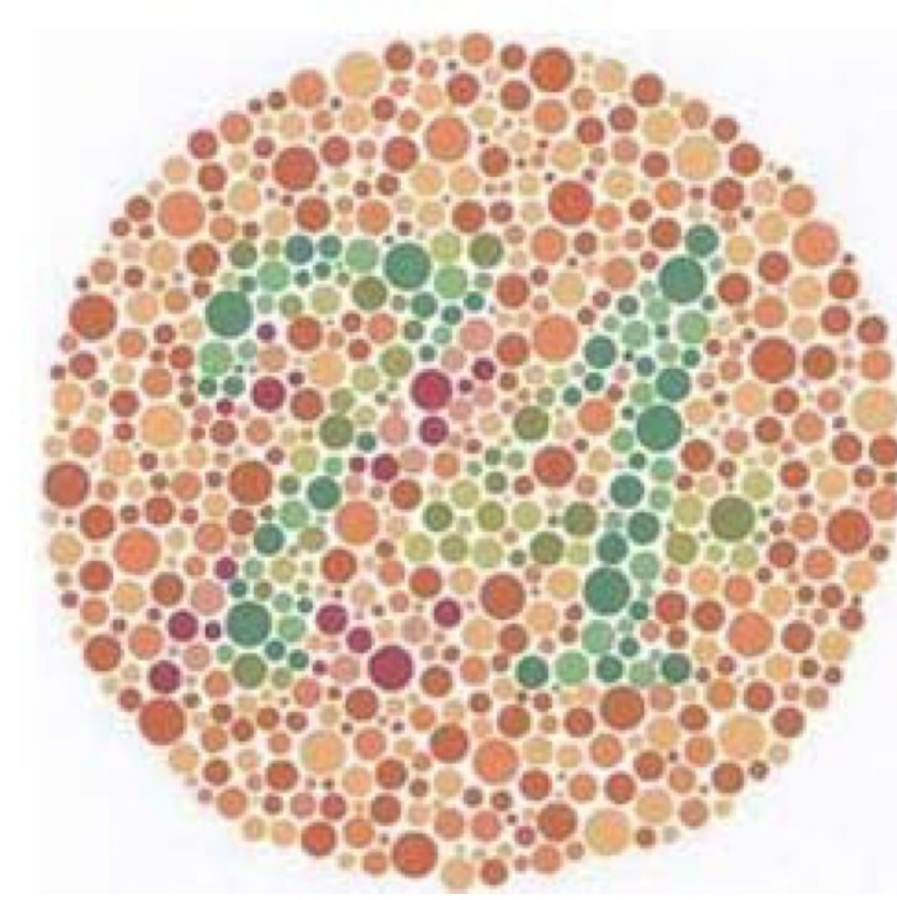
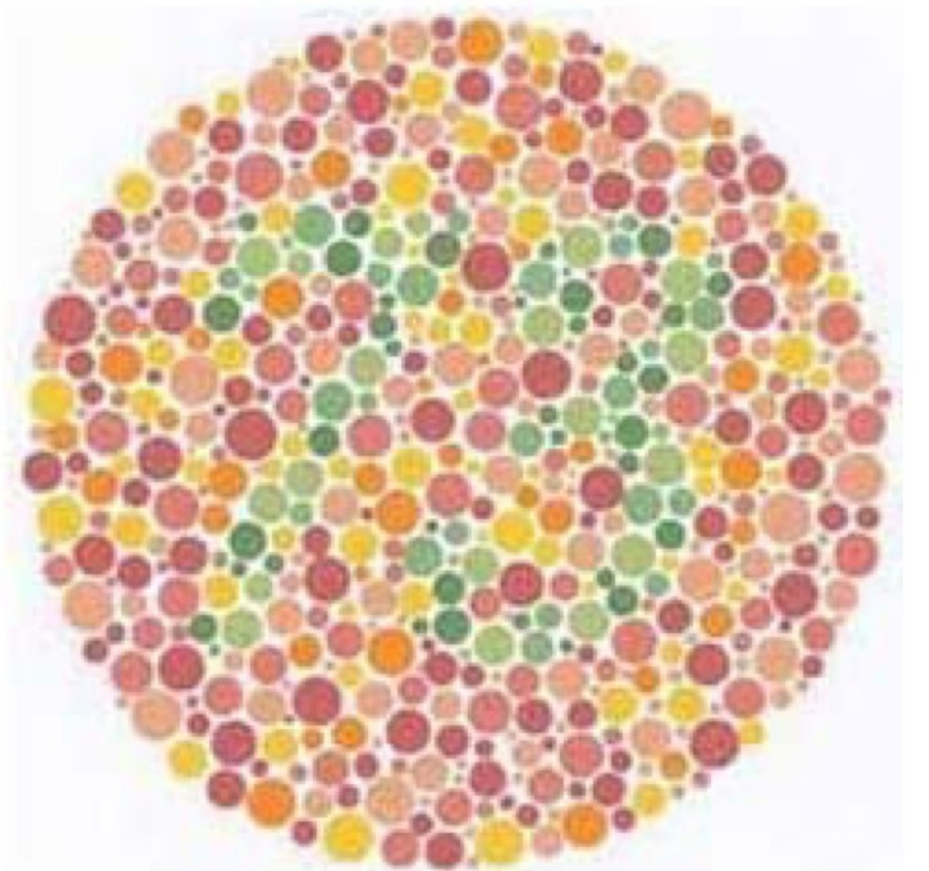
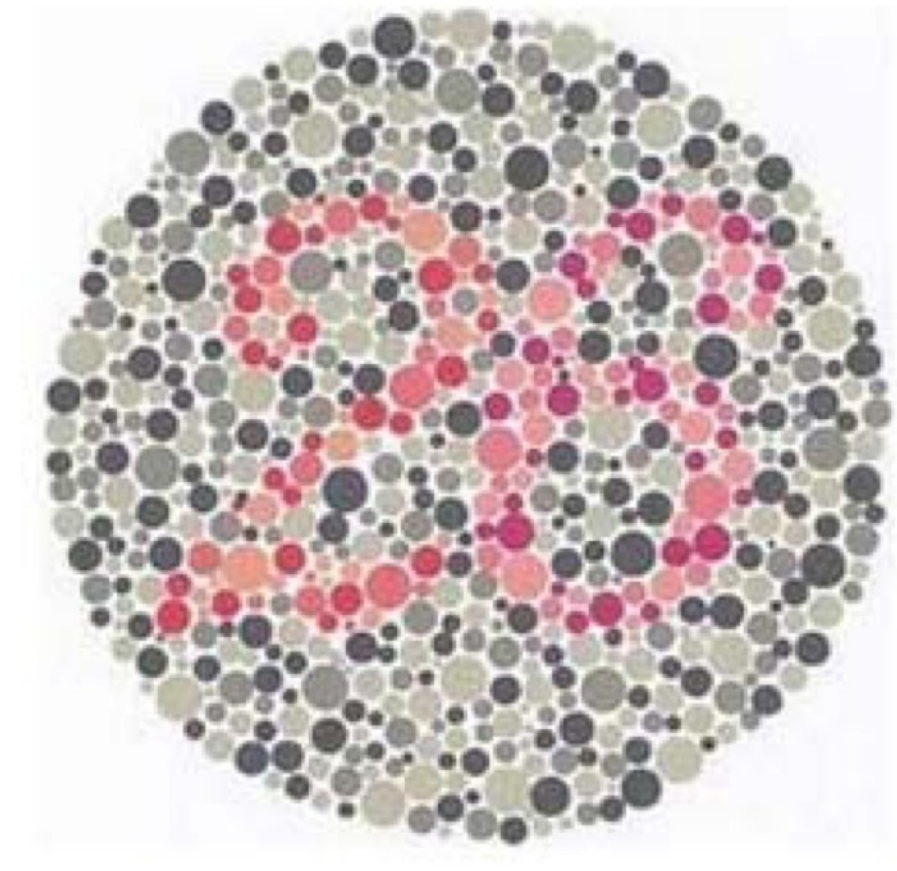
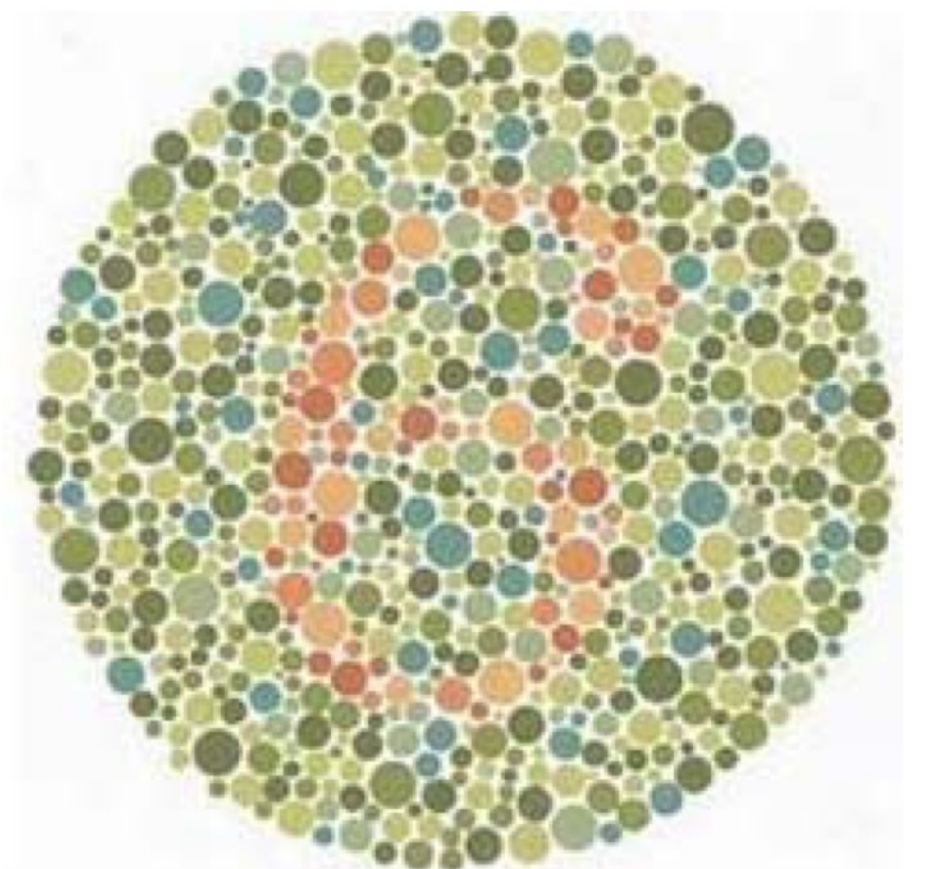
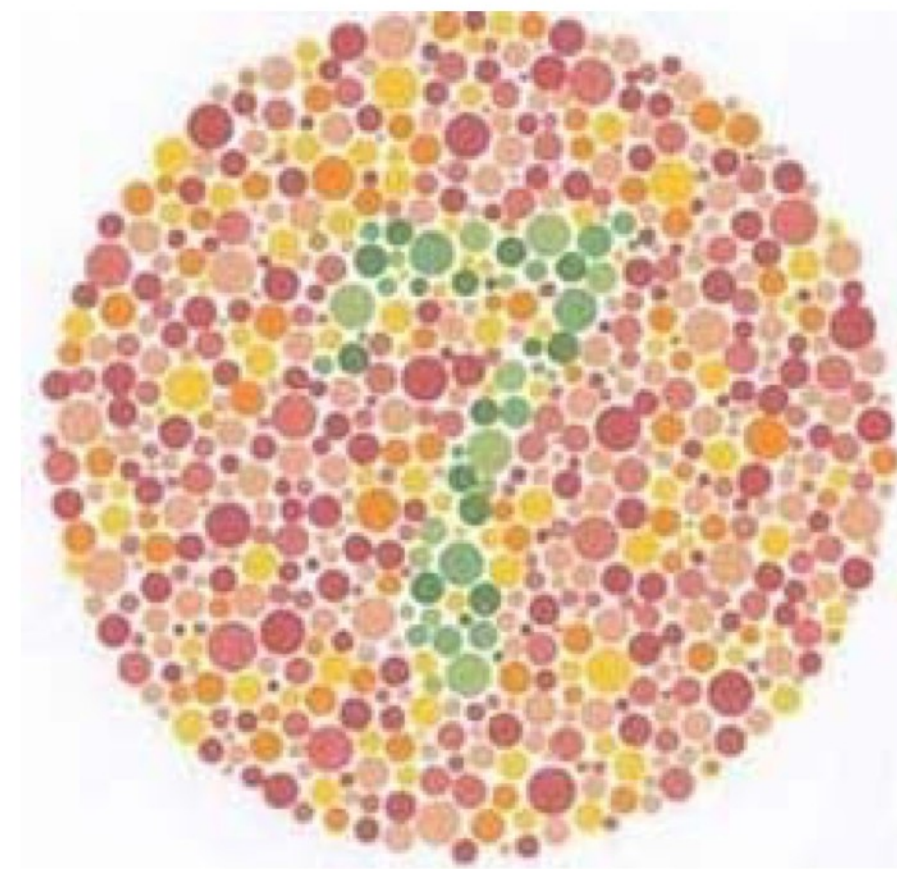
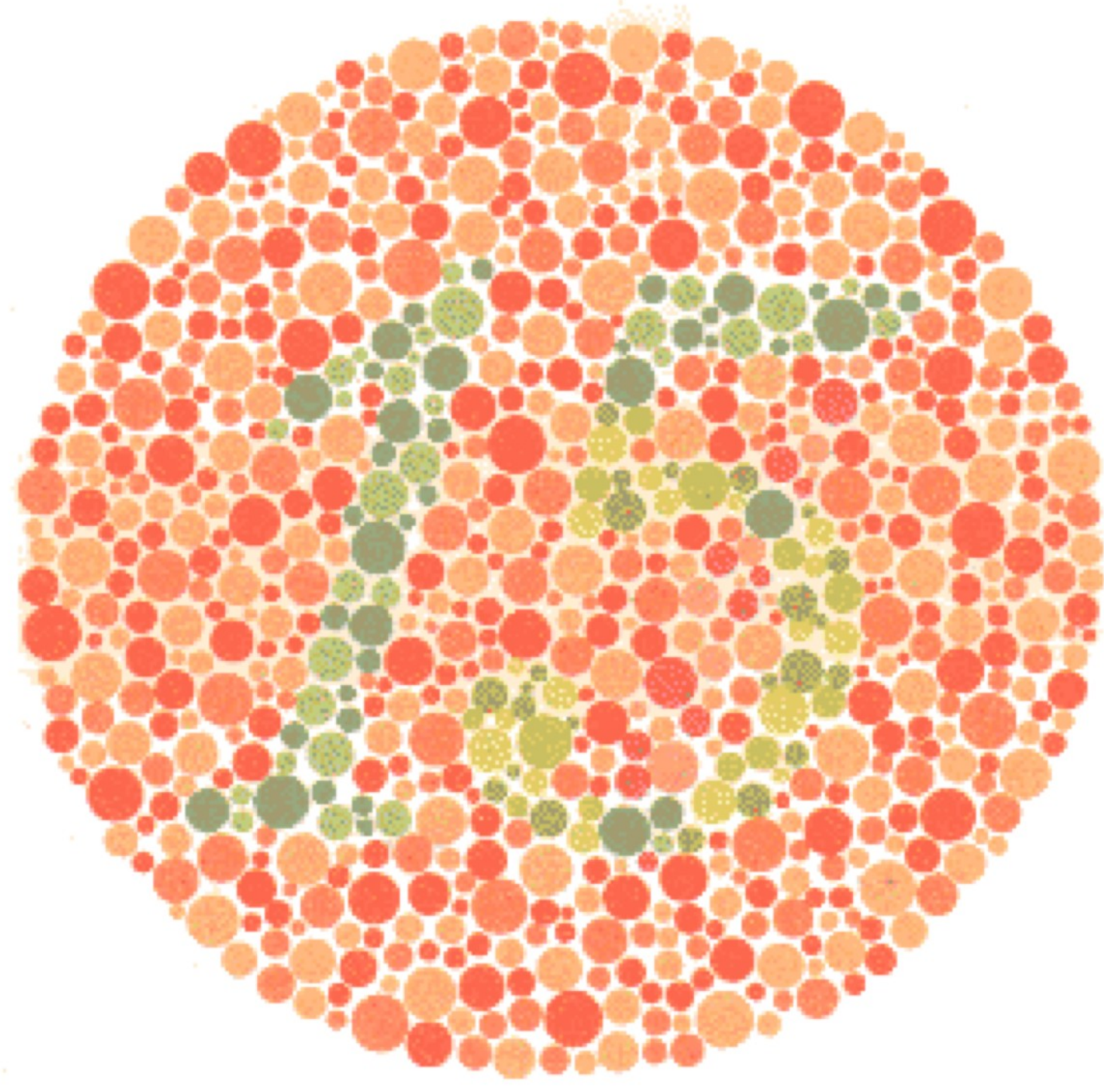
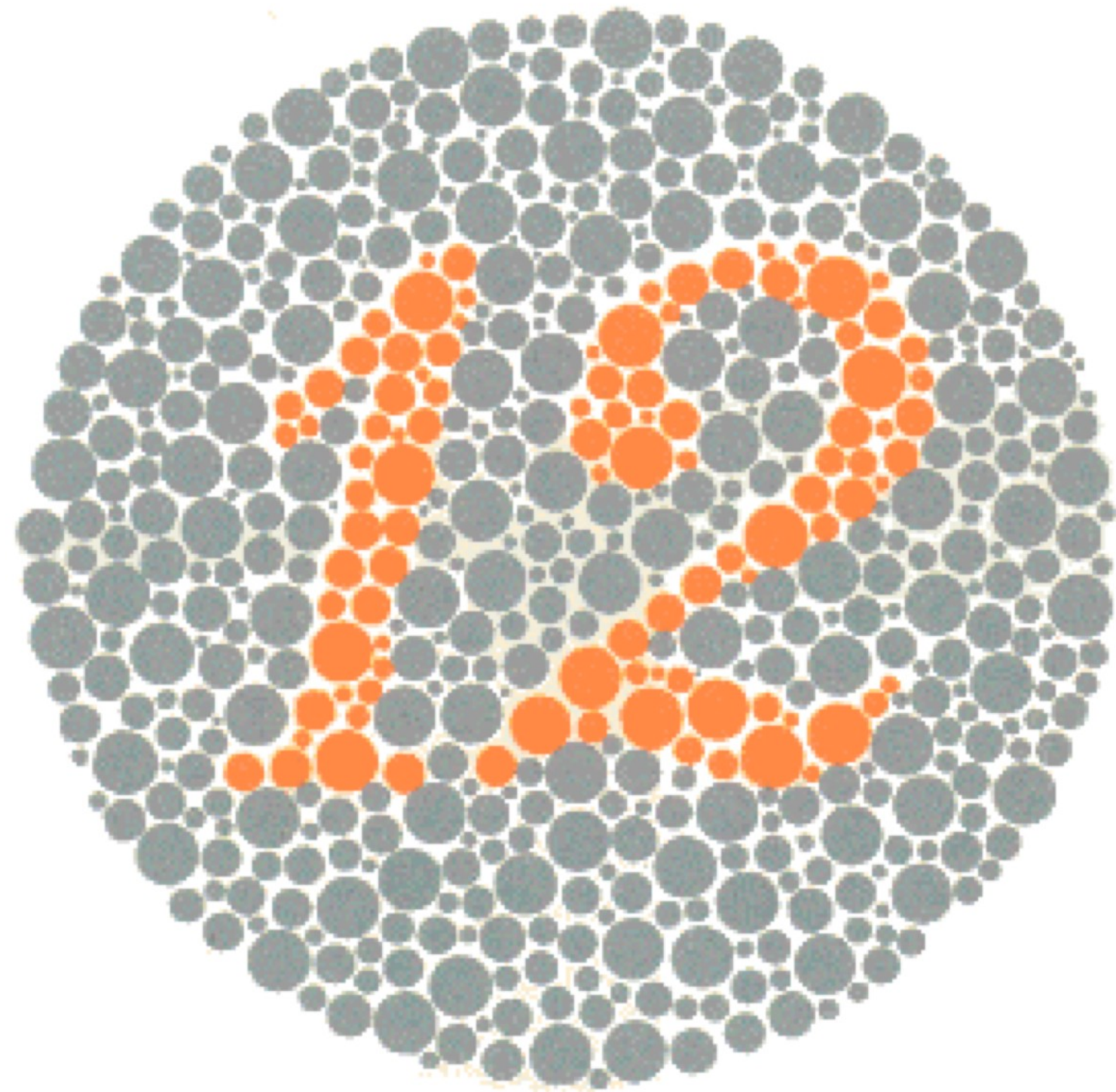
Deuteranopia (no green receptors)



Protanopia (no red receptors)



COLOR BLINDNESS TESTS



All Spending

Types of Spending

Changes

Department Totals

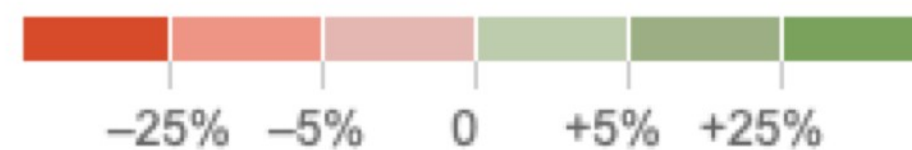
How \$3.7 Trillion Is Spent

Mr. Obama's budget proposal includes \$3.7 trillion in spending in 2013, and forecasts a \$901 billion deficit.

Circles are sized according to the proposed spending.

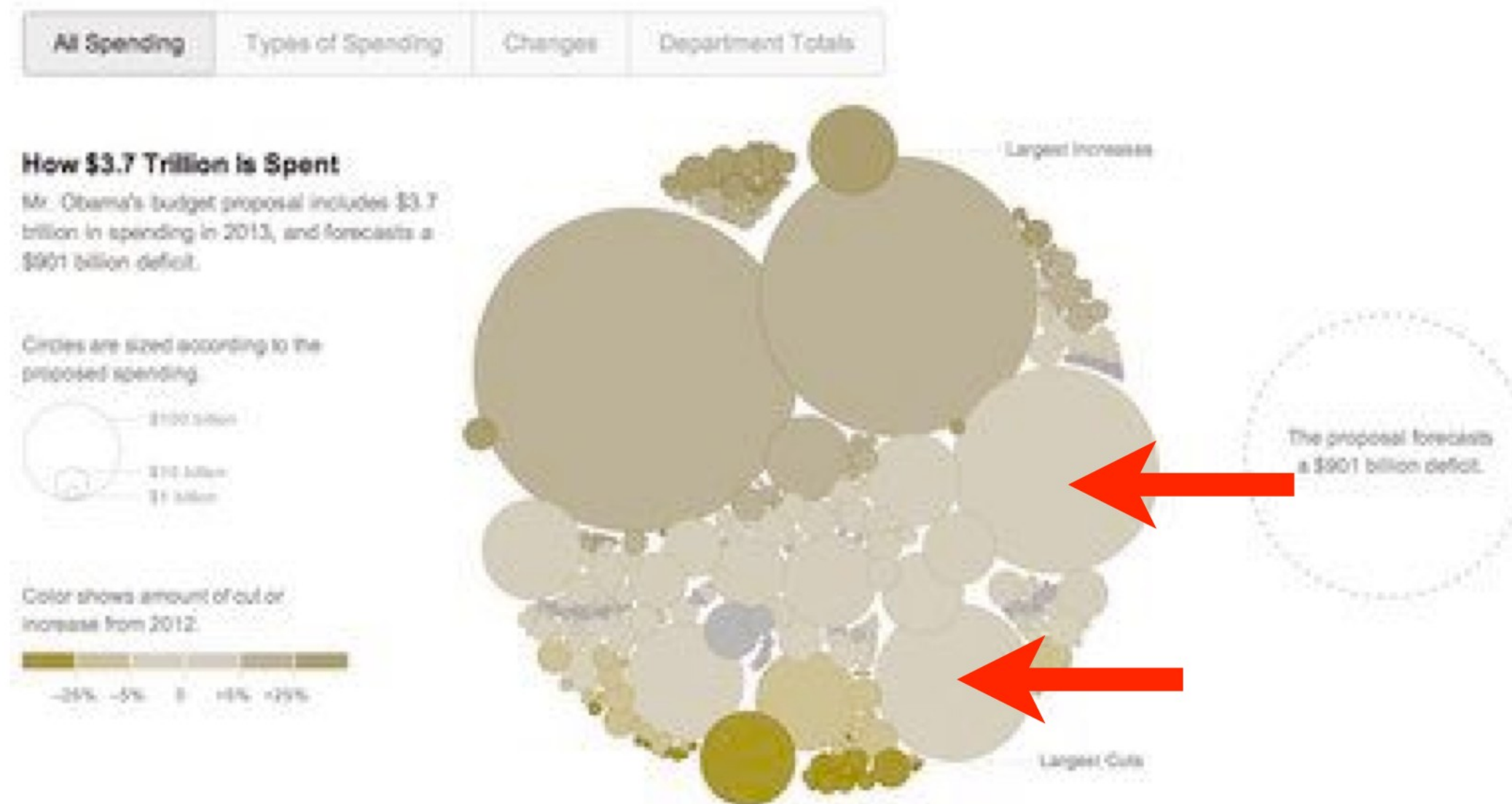
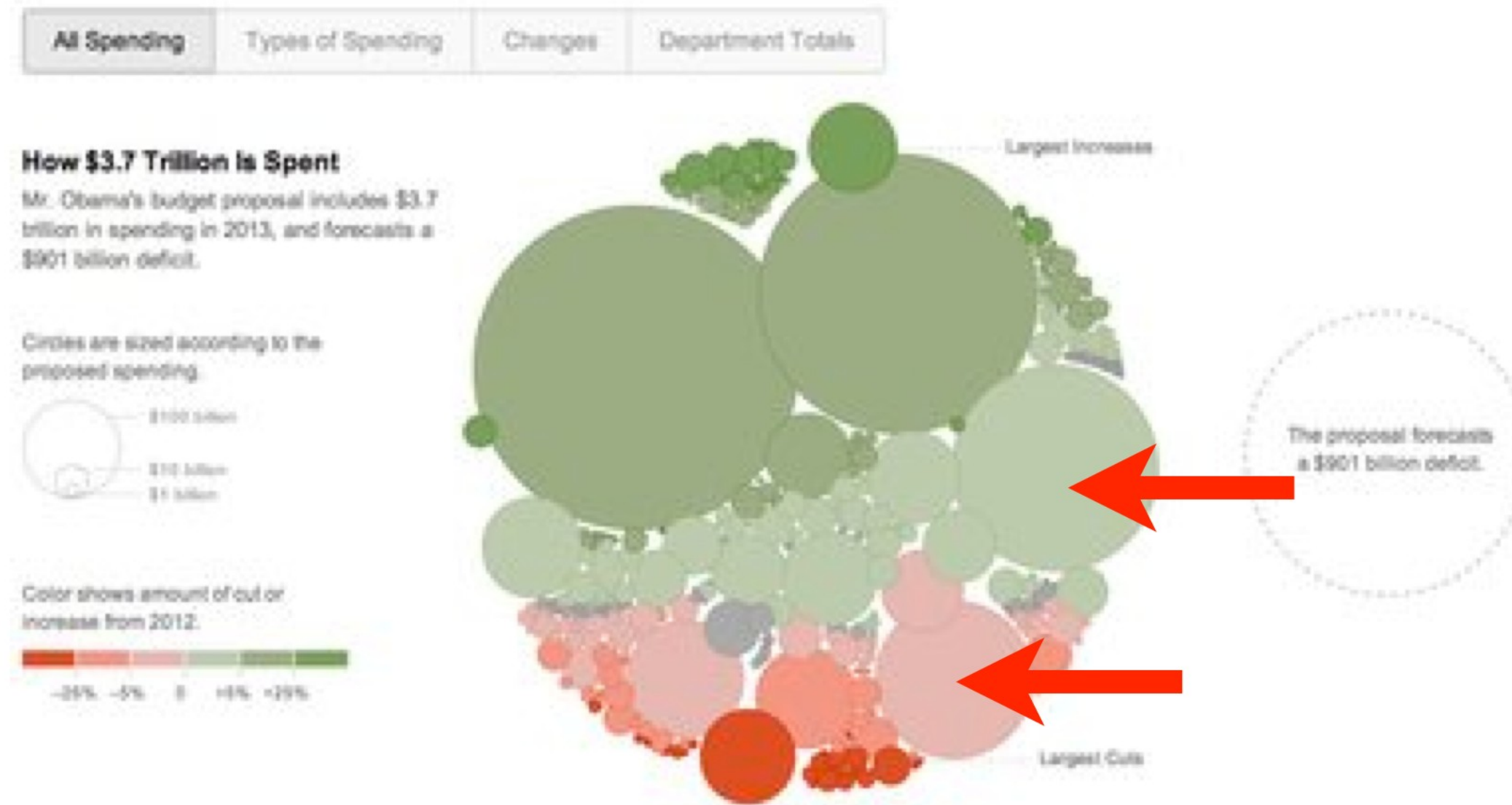


Color shows amount of cut or increase from 2012.



The proposal forecasts a \$901 billion deficit.

Source: bit.ly/budget-vis



GAME: PLACE THE COLORS IN ORDER

A

B

C

D

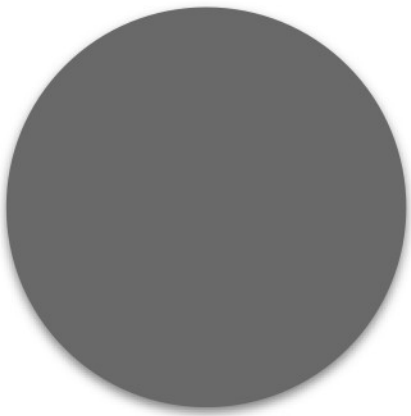
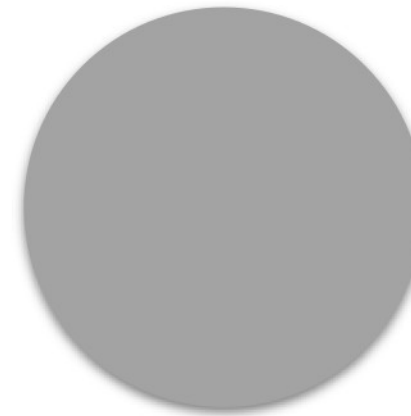
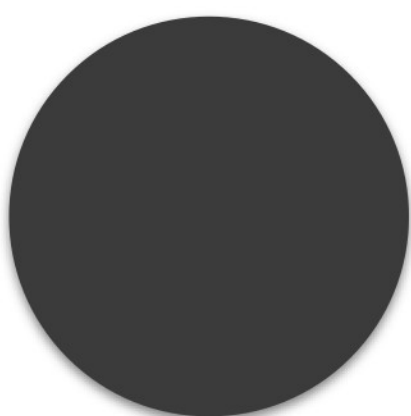
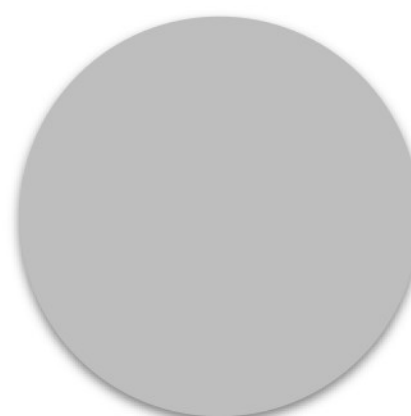
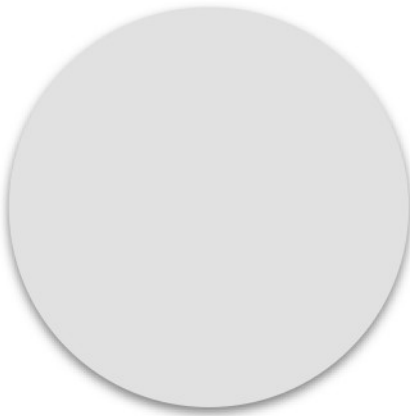
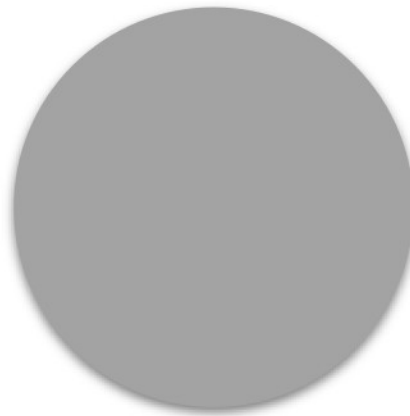
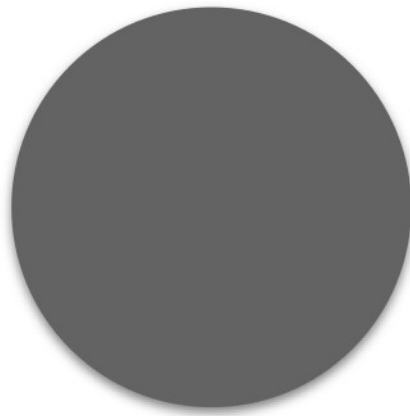
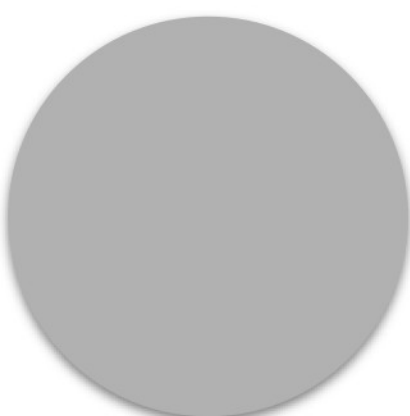
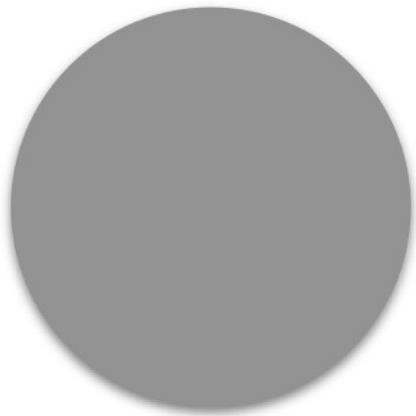
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GAME: PLACE THE COLORS IN ORDER

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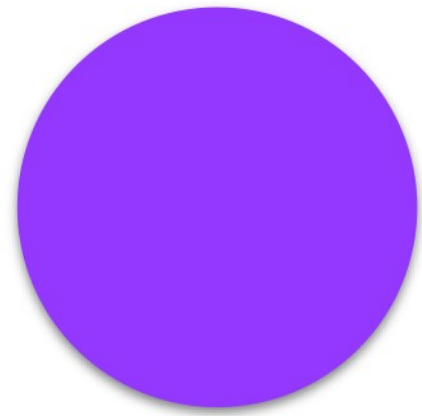
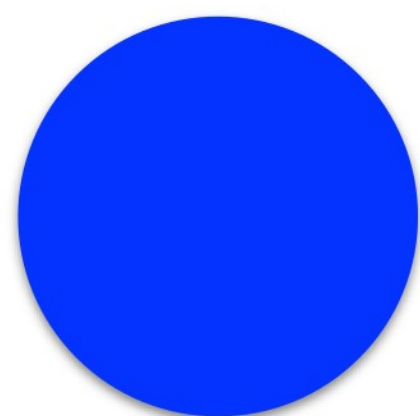
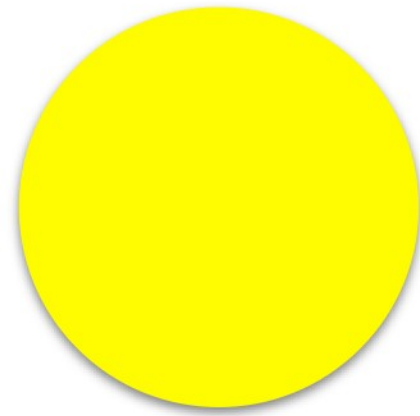
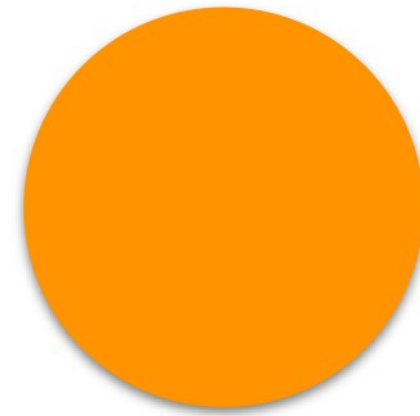
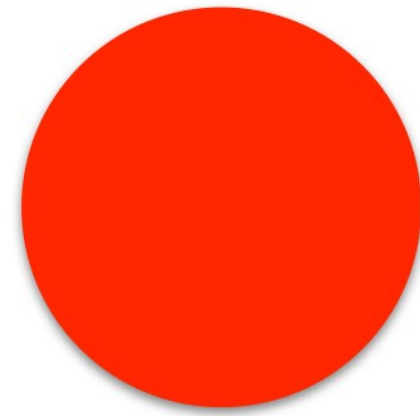
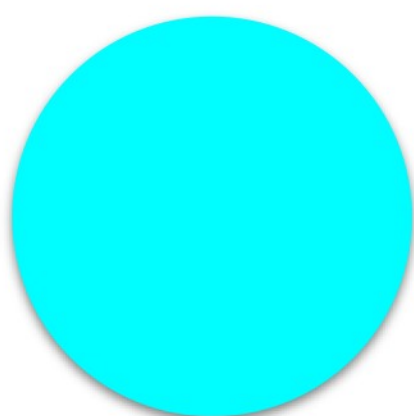
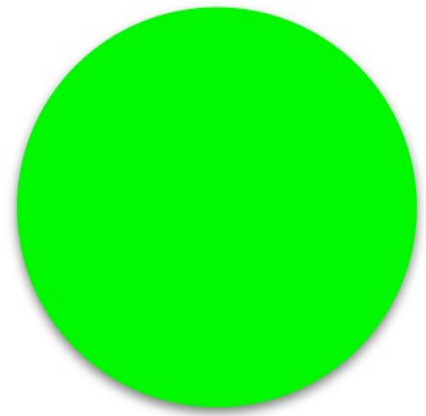
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GAME: PLACE THE COLORS IN ORDER

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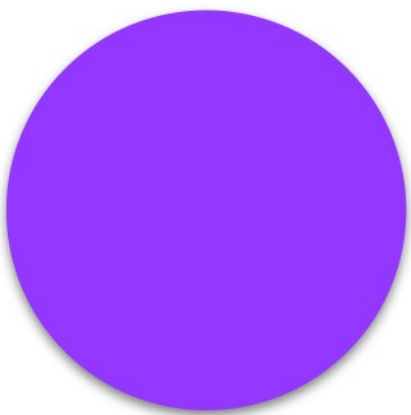
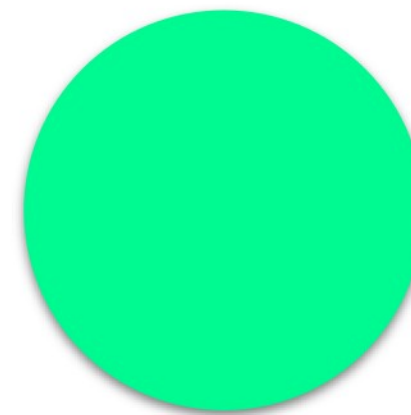
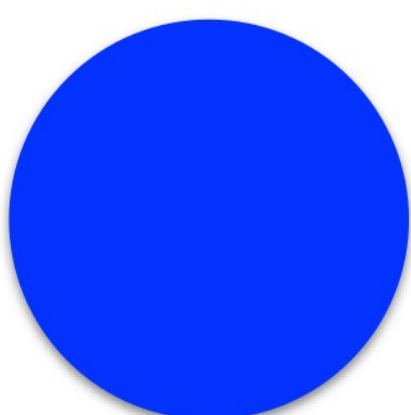
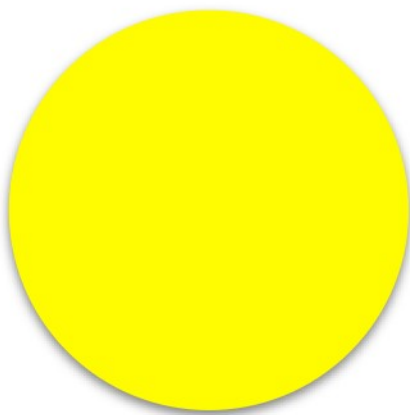
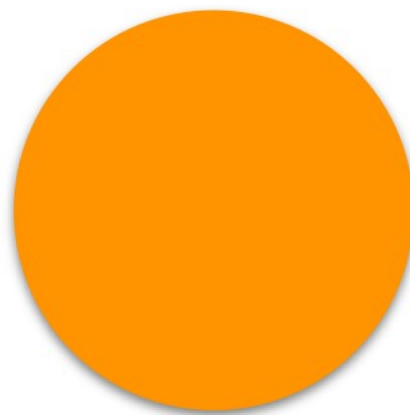
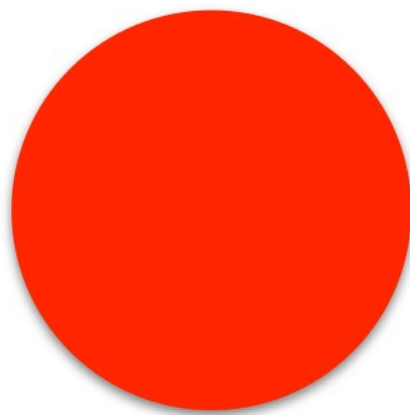
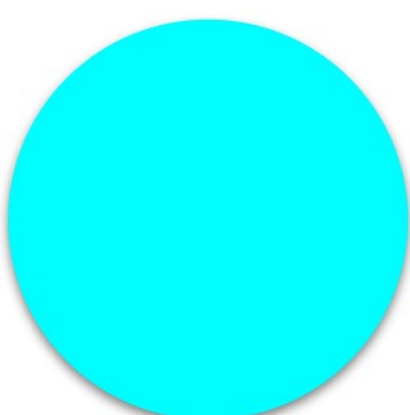
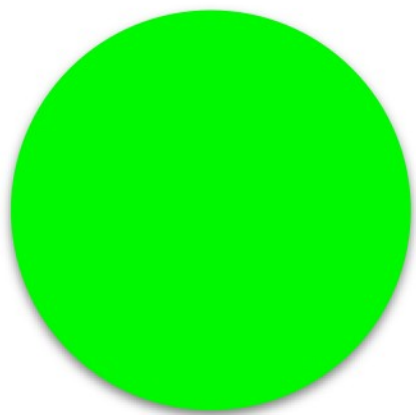
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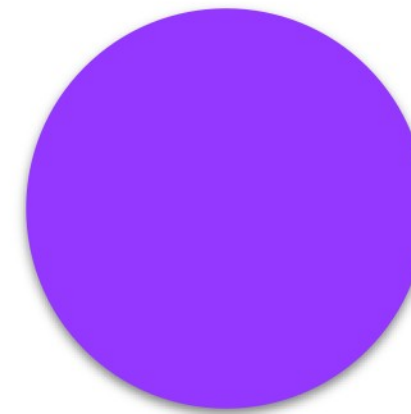
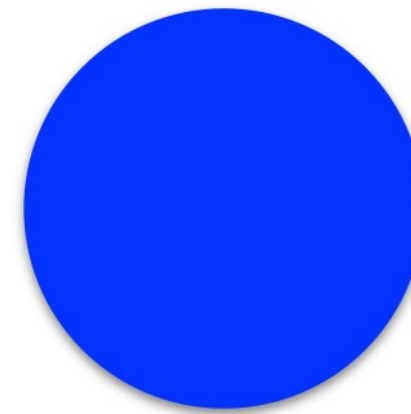
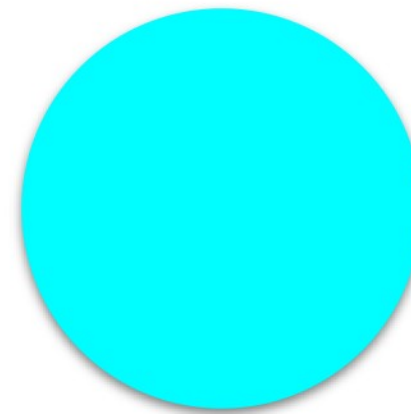
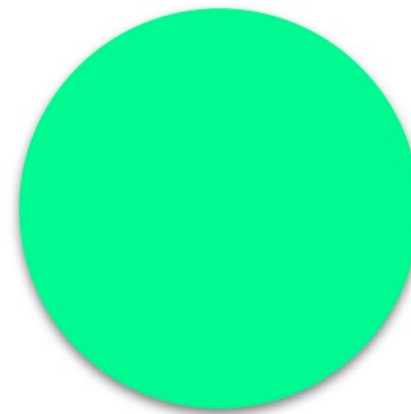
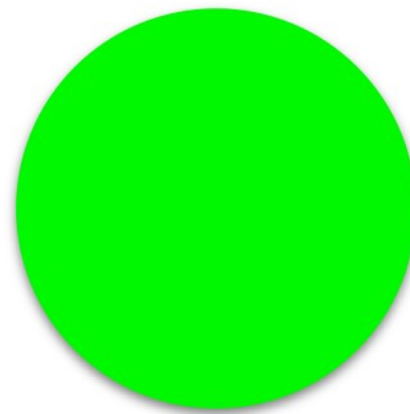
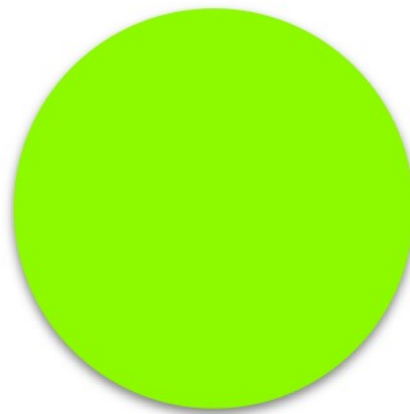
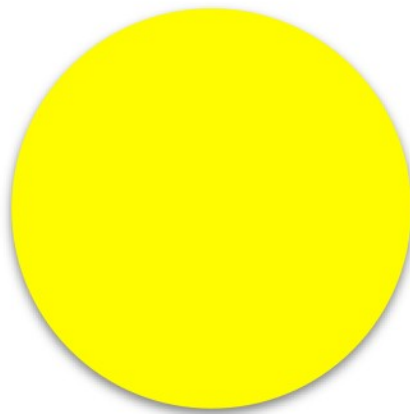
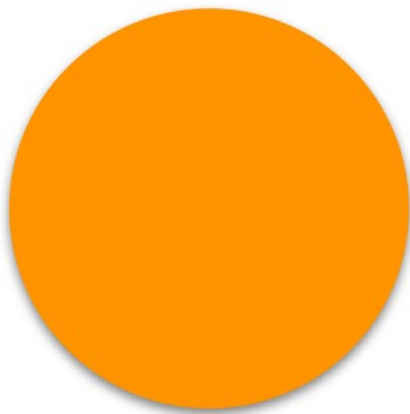
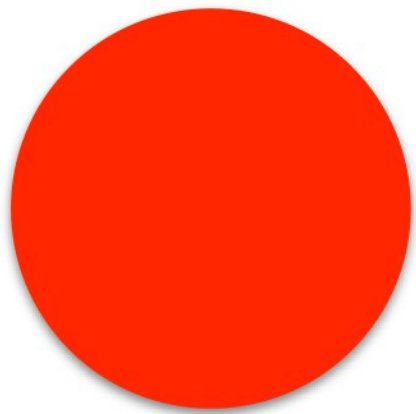
A

H

B

G

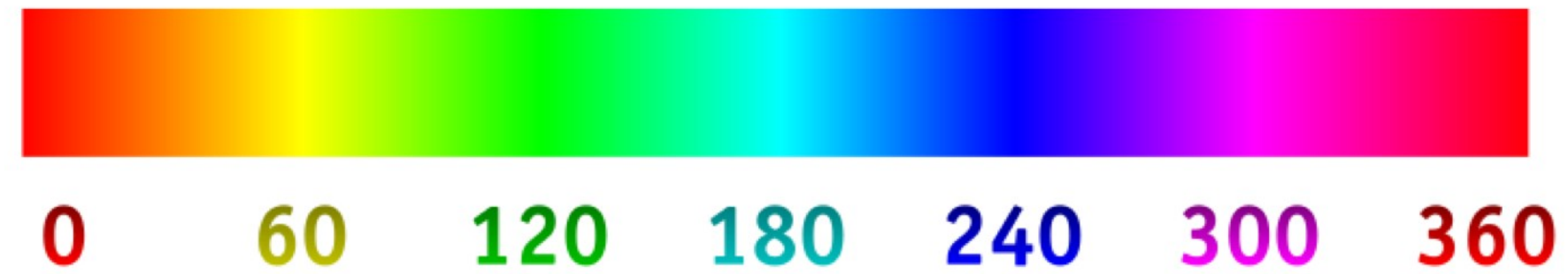
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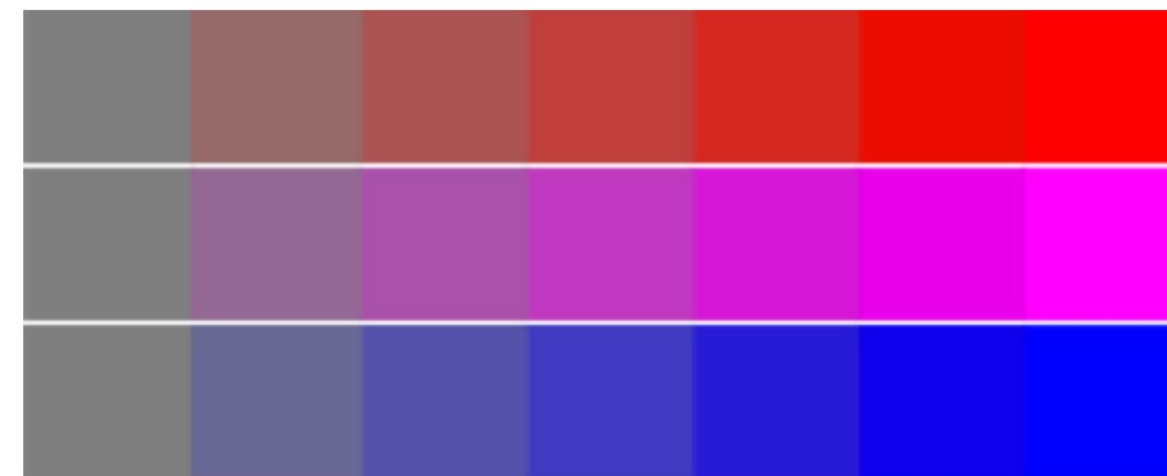
Hue isn't always reliable, so we
should make sure the colormap
still makes sense without it:
luminance, saturation

DIMENSIONS OF COLOR

Hue



Saturation



Different Saturation, medium Value

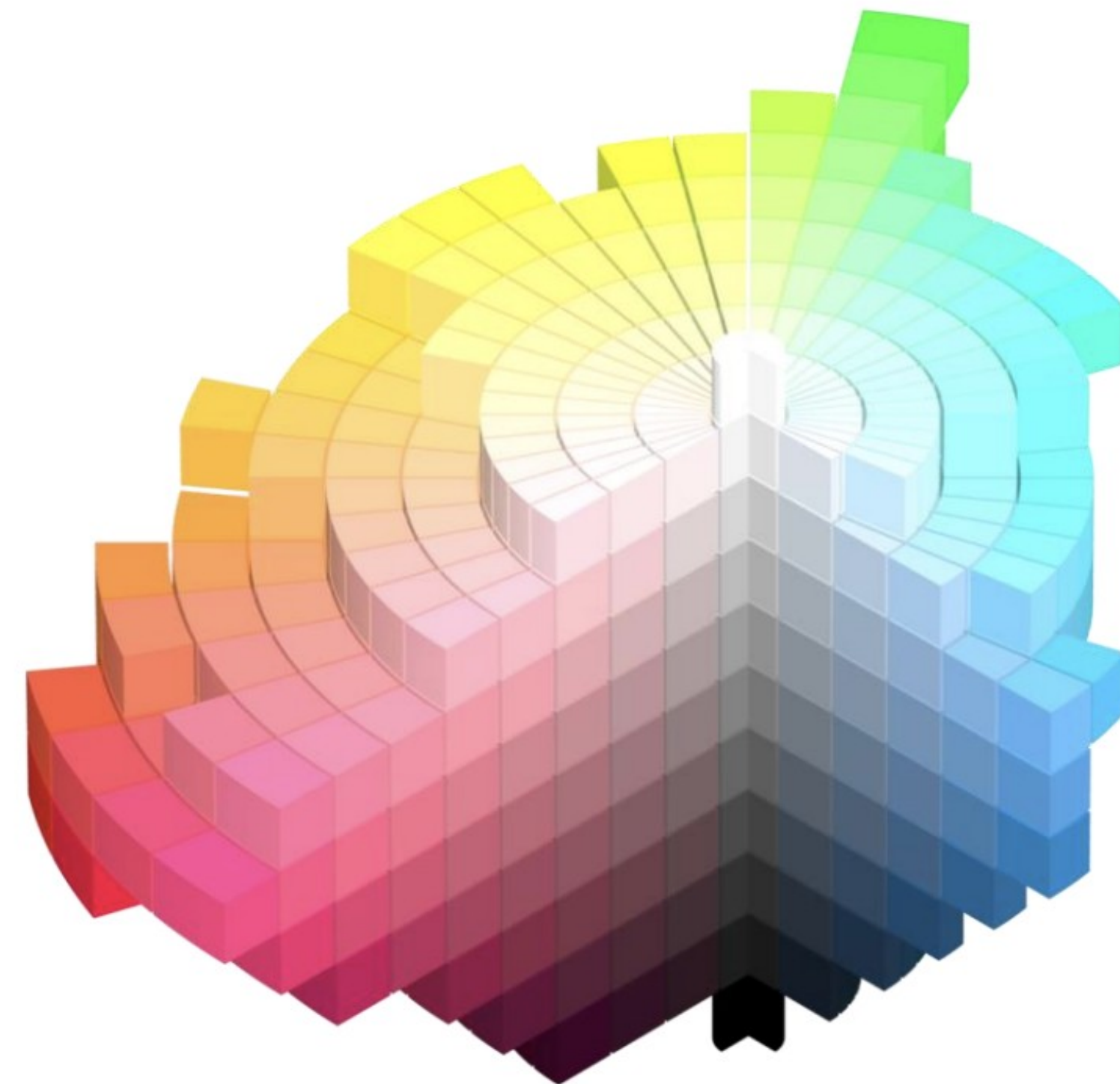
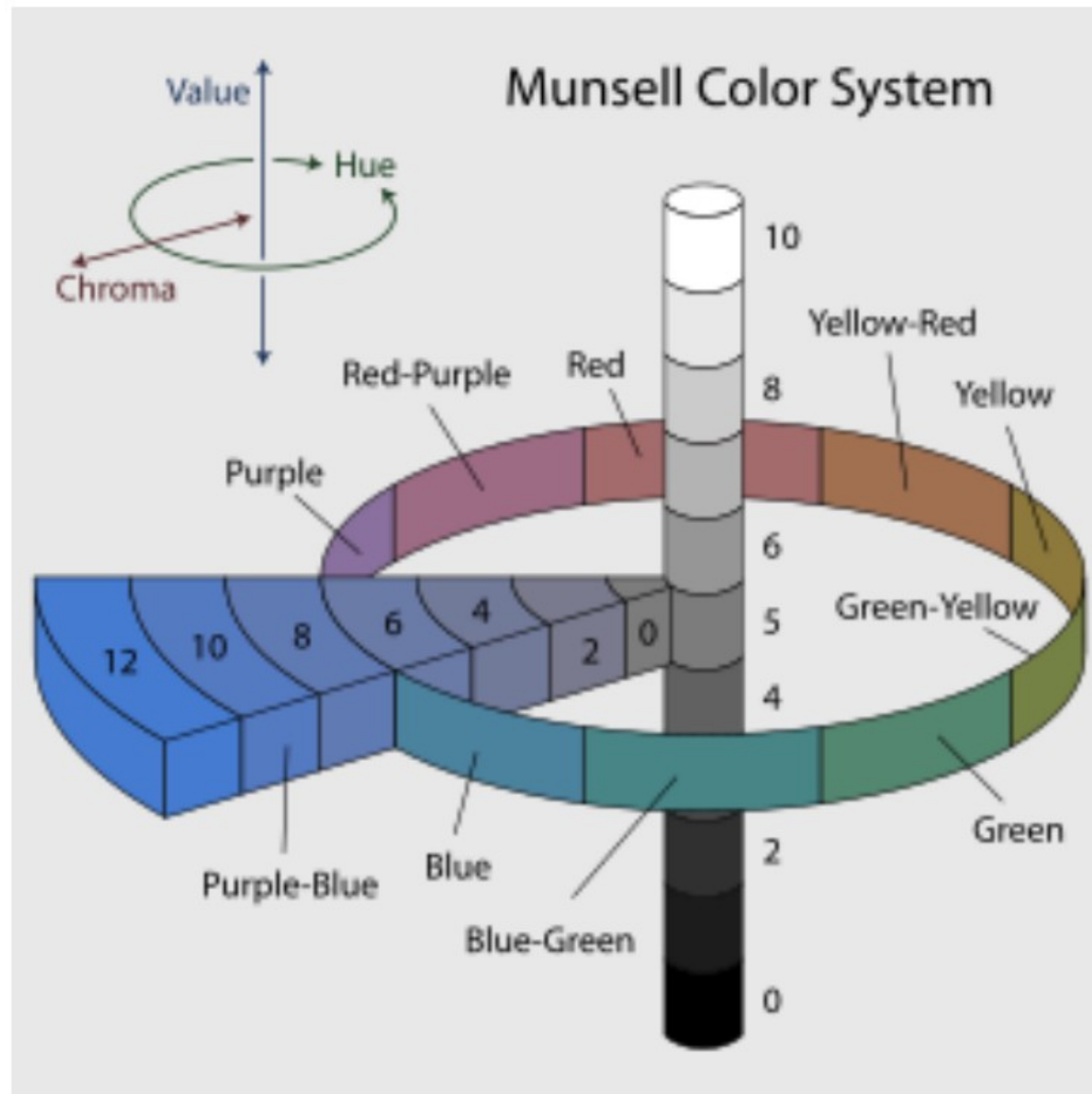
Luminance



Different Values, low Saturation

- Color space is 3D

- How to specify a given color in **perceptual** terms?
- Albert Munsell (~1930): hue, chroma, lightness (HCL)
- These form **perceptually uniform** & **independent** dimensions

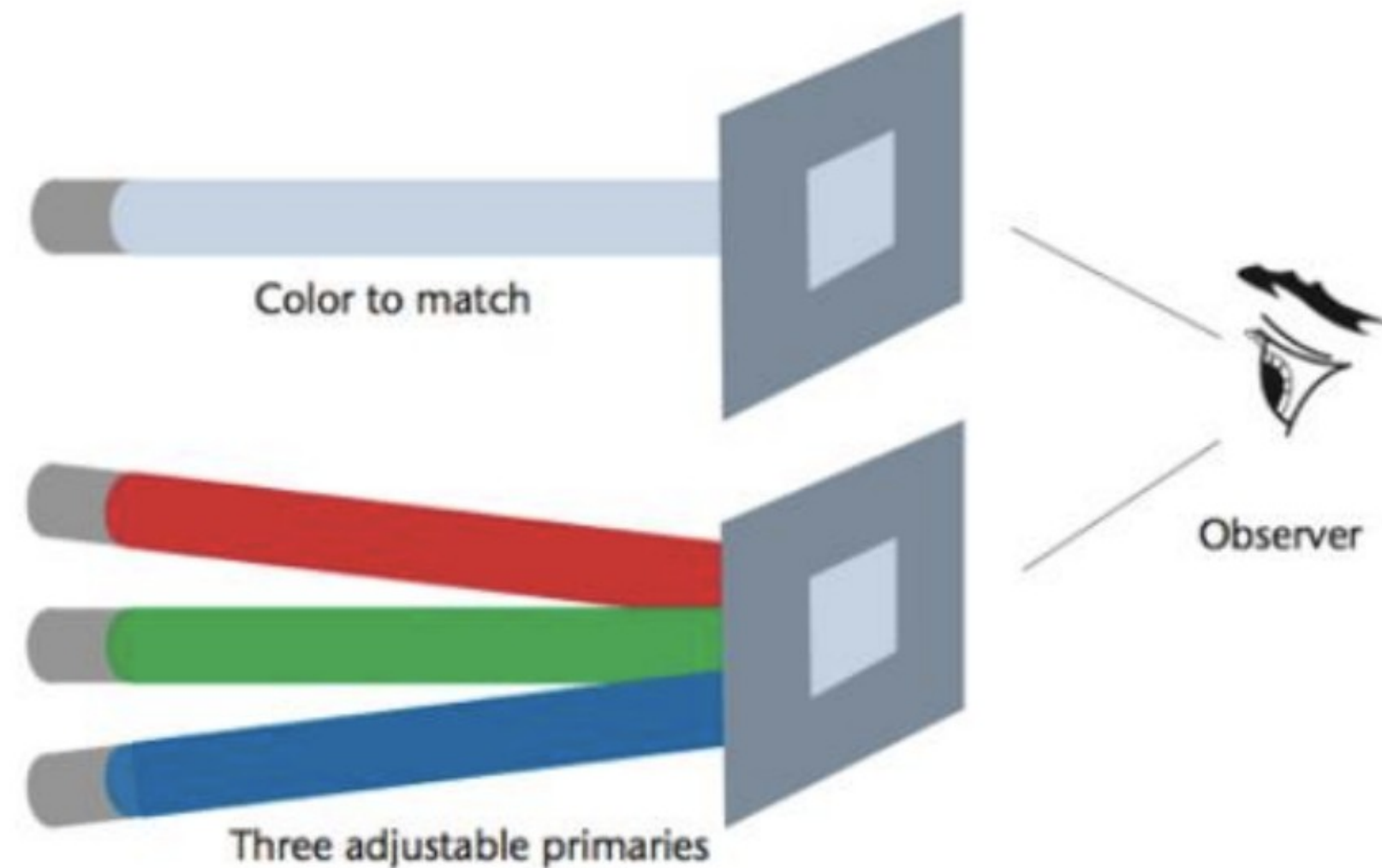


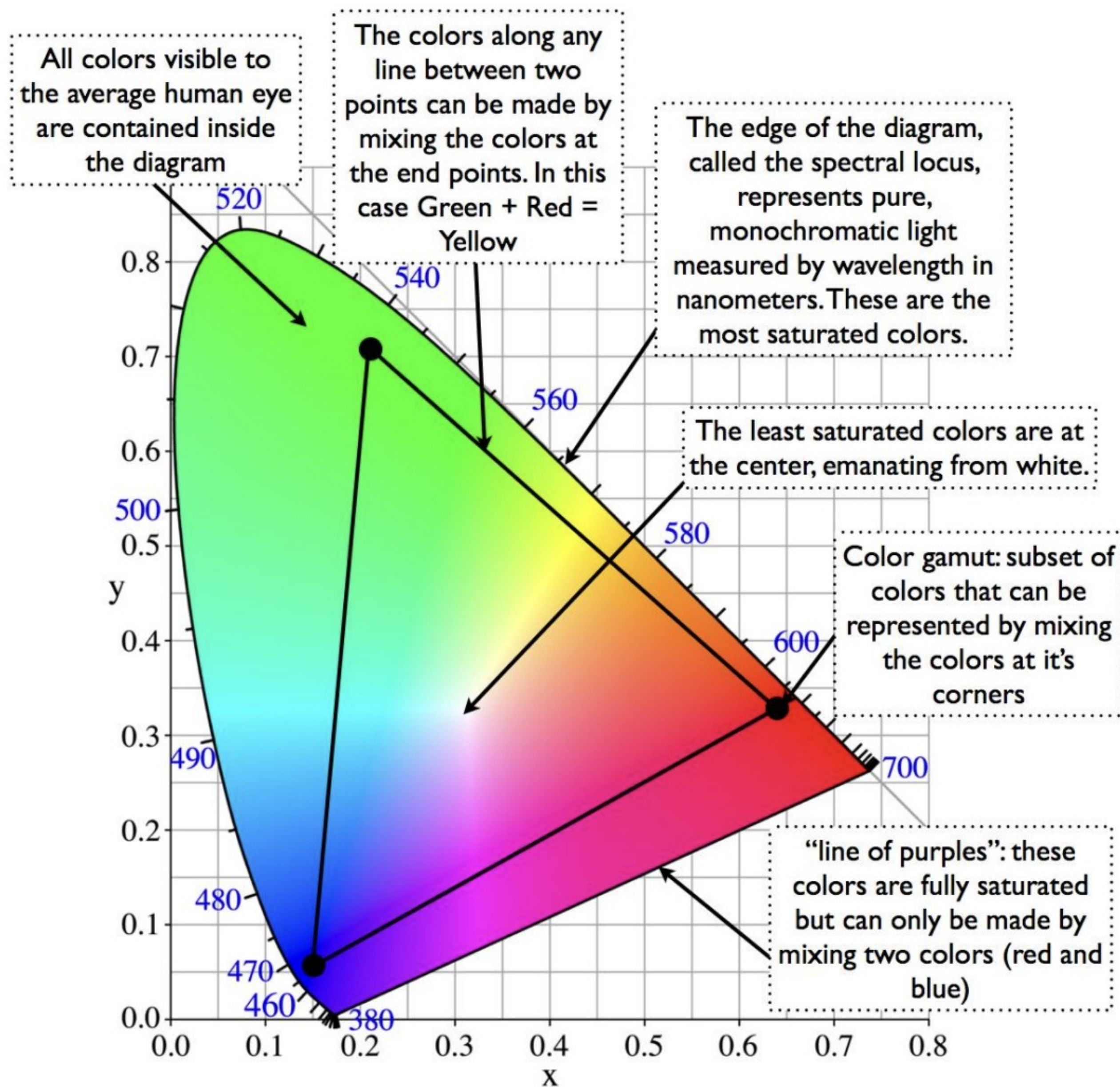
CIE Color space

- How do we know about the **perceptual** properties of colors, taking spectral sensitivity into account?

Experiments used a **color-matching** task:

- Adjust the intensity of pure R, G, B lights to match a given color
- This defines a new color theory connecting **physical** properties and human **perception** (spectral sensitivity).
- The CIE (International Commission on Illumination) becomes the standard to calibrate color in scientific instruments and human experiments



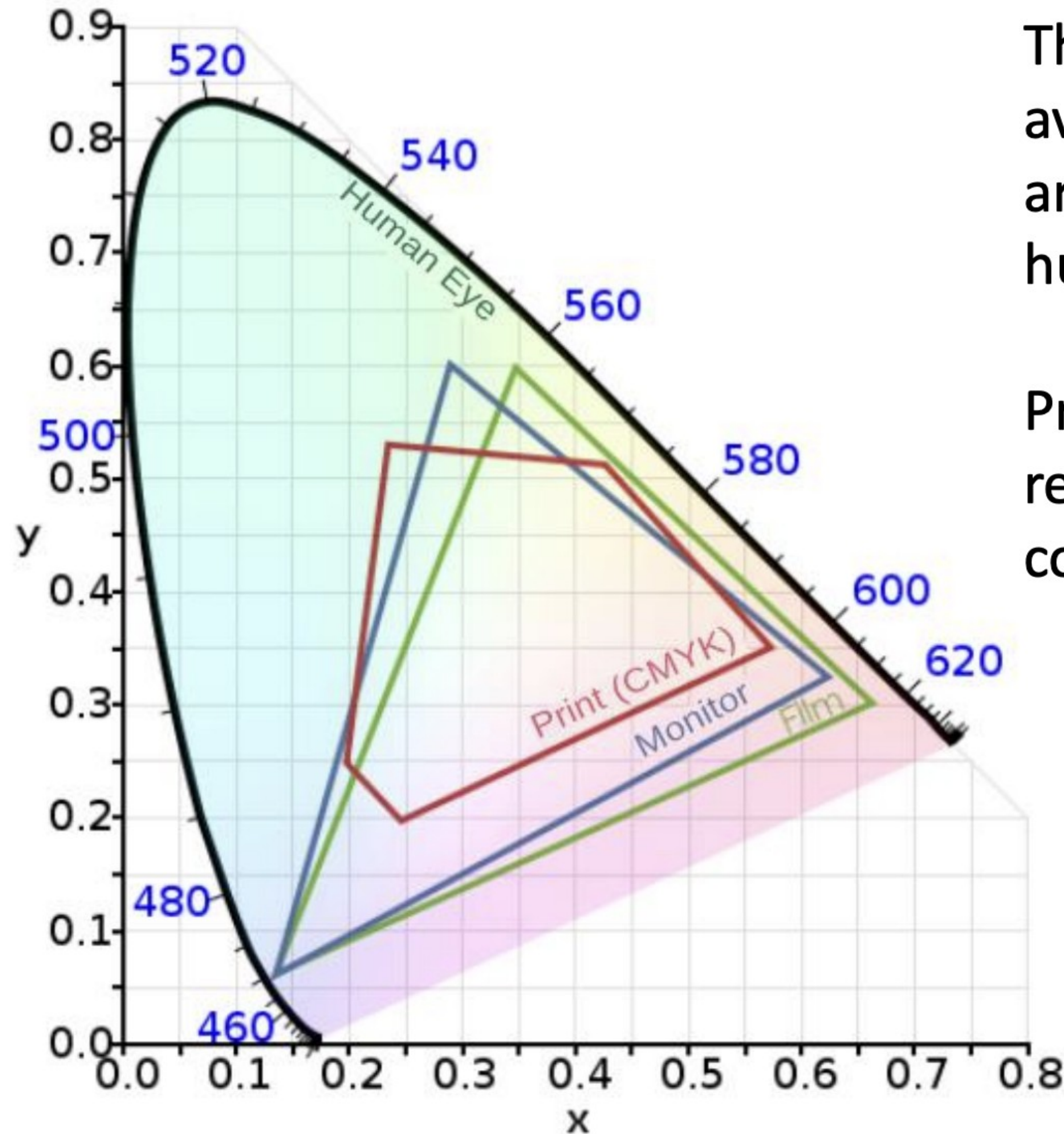


Anatomy of a CIE Chromaticity Diagram

The International Commission on Illumination (CIE) in 1931 defines a color space of (x,y,z) coordinates based on color-matching experiments combining **R, G, B** light sources in additive mixtures, and a “standard colorimetric observer”

This defines a new color theory connecting **physical** properties and human **perception** (spectral sensitivity).

There are eventually a variety of CIE color spaces (CIElab, CIEluv, ...) and lots of formulas for converting among them.



The standard gamut of colors available for different **display media** are a restricted subset of what the human eye can see.

Print (CMYK) is most restricted, and requires a more careful choice of color in graphics

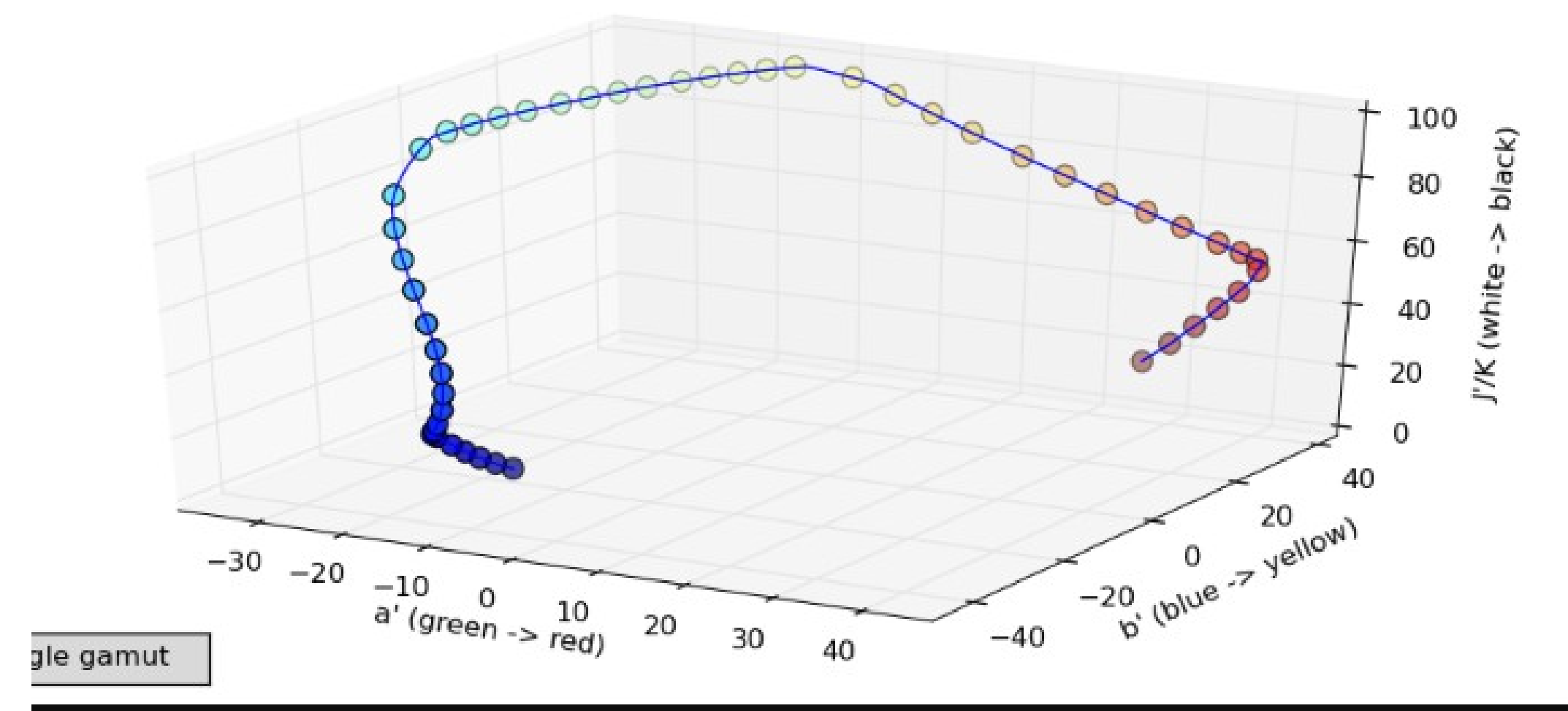
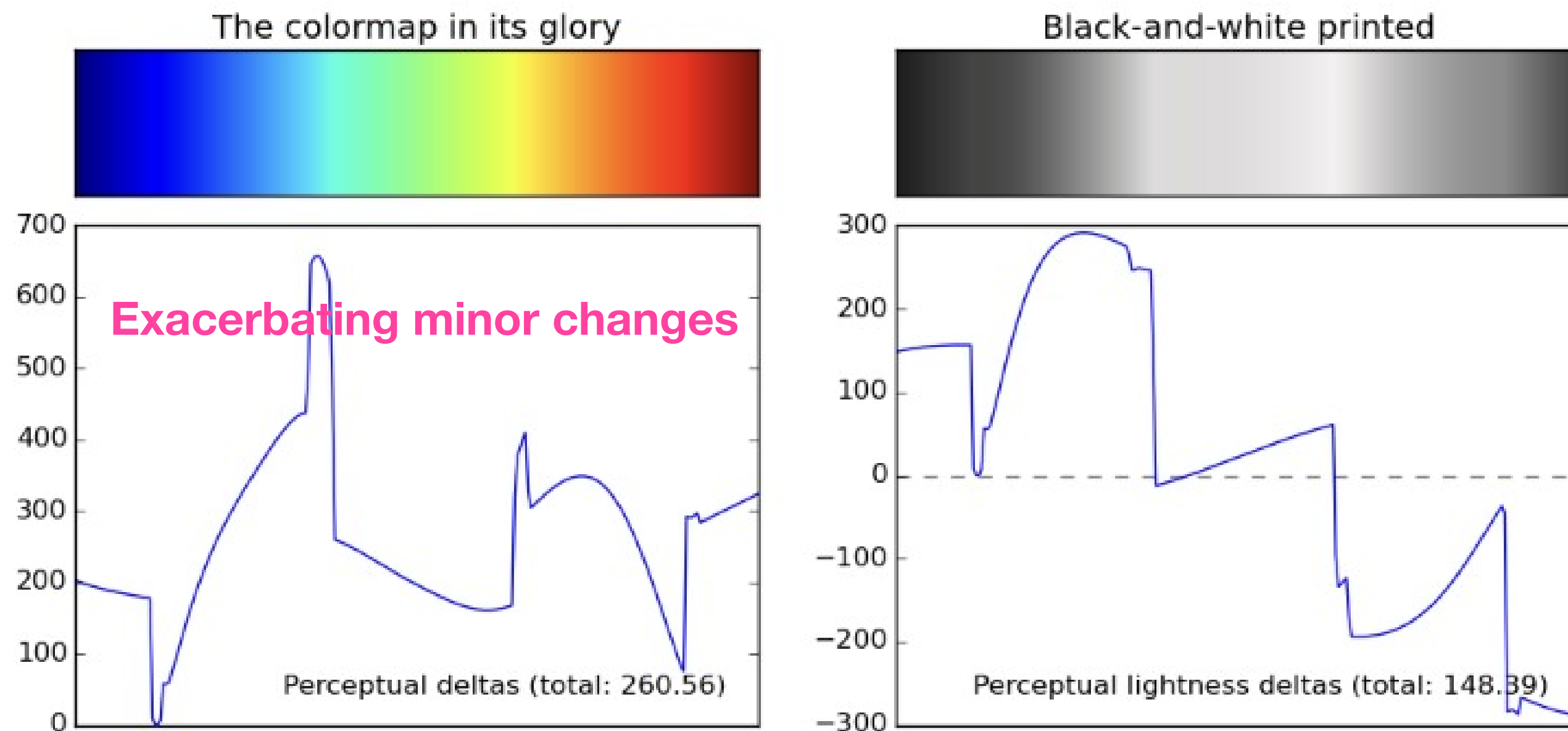


Always check the display on different media.

Goal: perceptually uniform colormaps through this space, that work in black and white

Effective and expressive use of color

jet in color space (one called CAM02-UCS)



“Perceptual delta” how much the perceived hue changes as you walk from one side of the colormap to the other

“Perceptual lightness deltas” how much the perceived lightness changes as you walk through the colormap

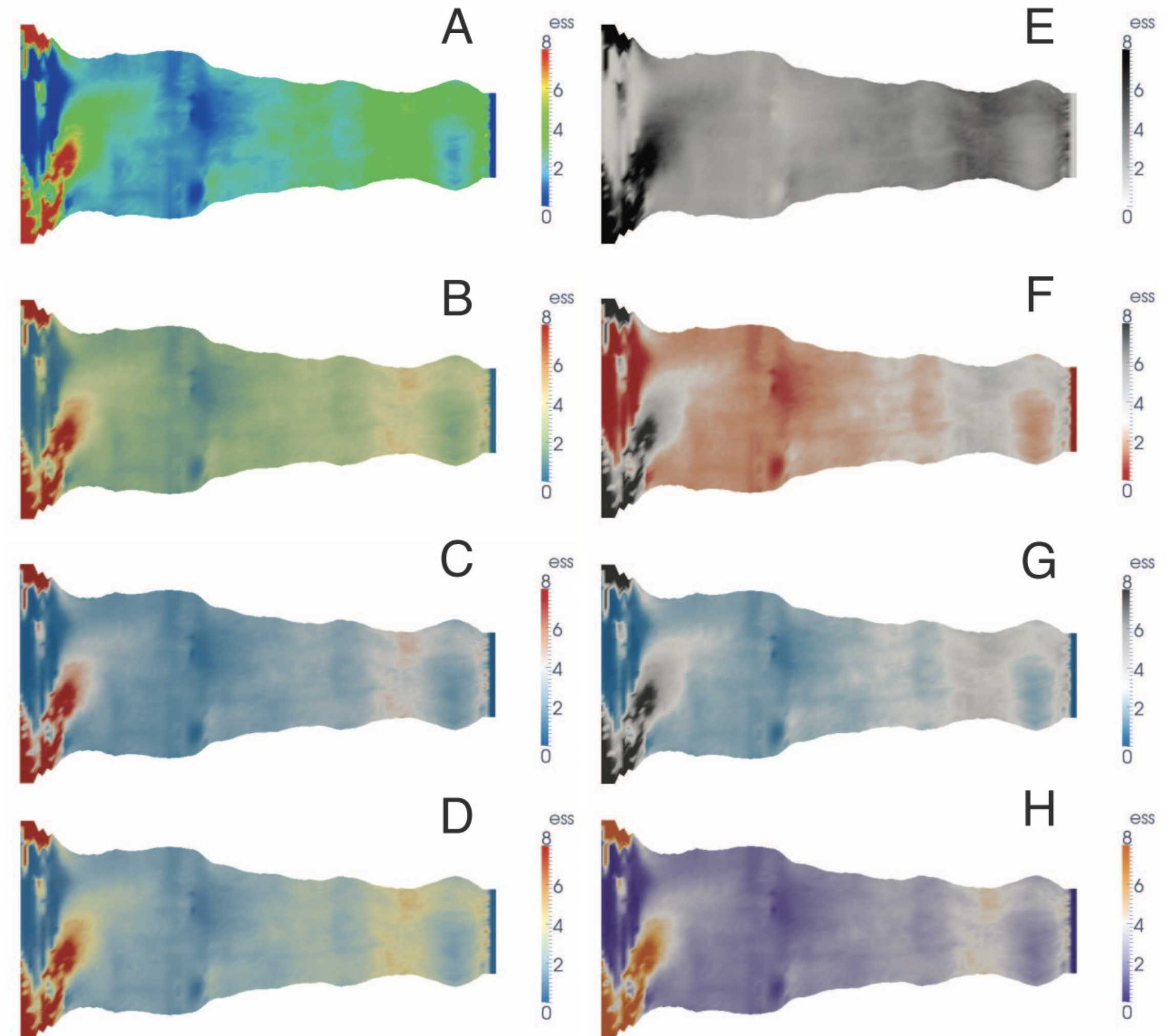
Design Choices Make a Difference

Borkin et al. Artery Visualization

You are a doctor who looks at cardiovascular imaging to diagnose heart disease.

The image shows a patient's artery, colored by Endothelial Shear Stress (stress on arterial walls), a risk factor for heart disease.

Which colormap do you think would make it easiest to identify problems in blood flow?

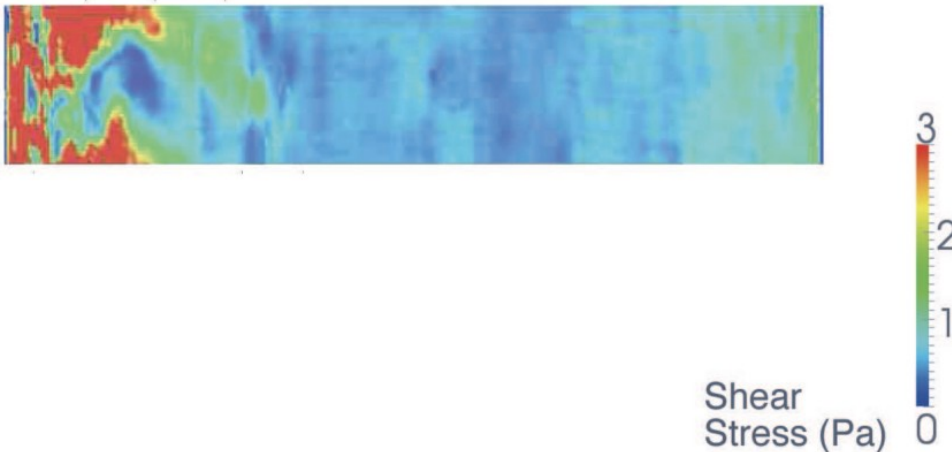


Design Choices Make a Difference

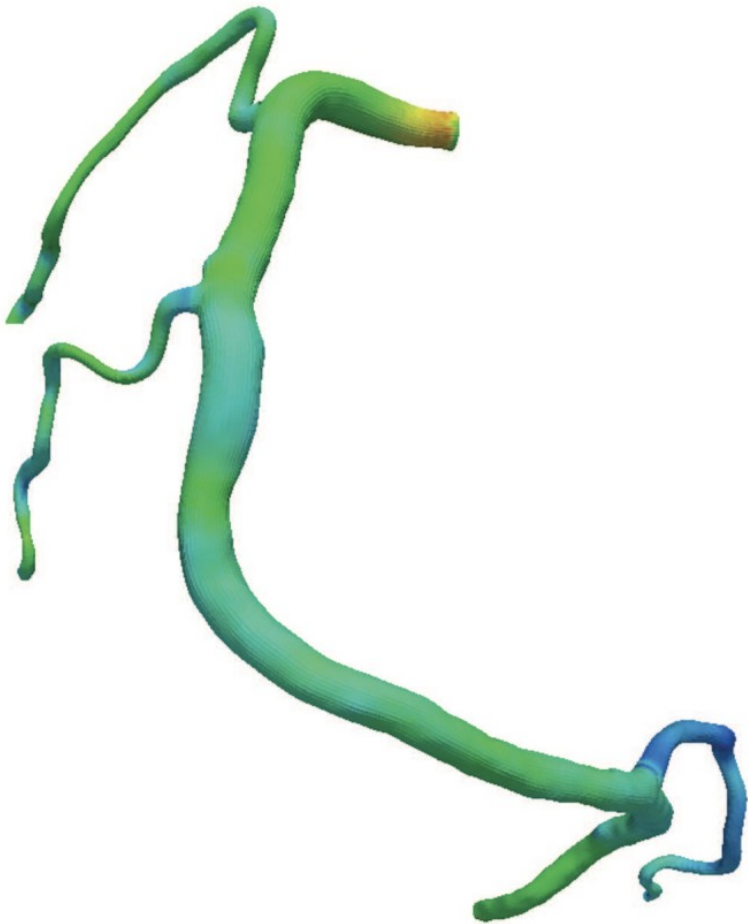
Borkin et al. Artery Visualization

Rainbow

62%



39%

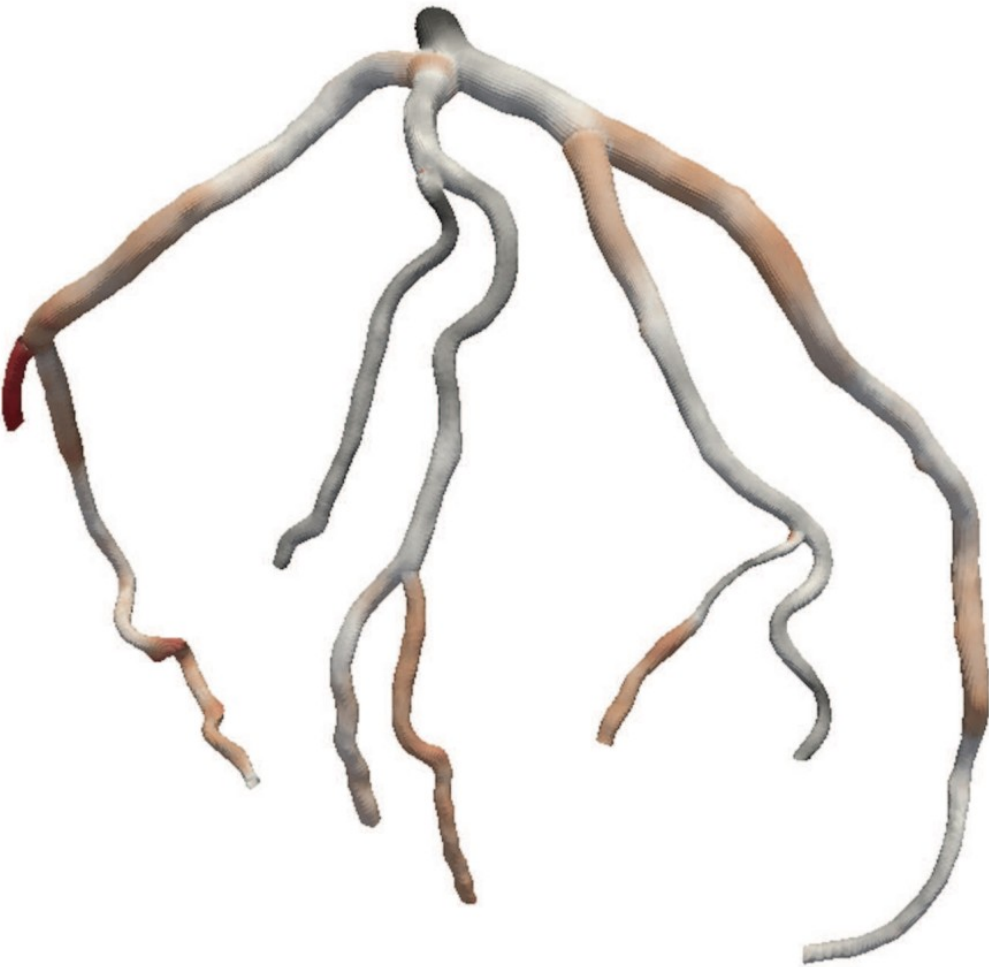


Diverging

91%



71%



Simple takeaways for color in graphics

- Ensure your colormap is *perceptually uniform*
- Ensure your colormap makes sense for color blind viewers
 - Use luminance and maybe saturation; hue isn't as reliable!
 - Check your colors in gray scale to make sure you can still interpret the encoding!
- Color (hue) should be used for nominal attributes

Make your own colormap

- Who can make the **best** one?
- Talk to your partner about the **common features** between your two best colormaps.
- Can you make a **better rainbow map**? What is the hardest criteria to improve?

**Colors for non-
quantitative data**